

THE HETEROGENEOUS EFFECTS OF ENTRY ON PRICES

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Abstract

We study the effect of entry on the price distribution in the German retail gasoline market. Exploiting several hundred entries over five years in an event study design, we find that entry causes a persistent first-order stochastic shift in the price distribution for at least three years after entry. Prices at the top of the distribution change moderately or not at all, but prices at the left tail decrease by up to 13% (9%) of stations' gross margins after entry within a 1 km (2 km) radius. The value of information (VOI) increases by 29% (15%) in 1 km (2 km) markets, suggesting larger savings for consumers with easy access to information. (JEL: D22, L11, L81, D83, R32)

The editor in charge of this paper was Pierre Dubois.

Acknowledgments: We would like to thank participants in the DICE Applied Micro Group, the DICE Brown Bag Seminar at the IIOC 2023, the MaCCI Summer Institute in Competition Policy 2023, the MaCCI Annual Conference 2024, and the Swiss IO Day 2024 for their valuable feedback. This project also benefited from discussions with and comments by Natalia Fabra, Mitsuru Igami, Andreas Lichter, Felix Montag, Mitsukuni Nishida, Andrea Pozzi, Chad Syverson, Stephanos Vlachos, Mo Xiao, and Biliiana Yontcheva. Funding from the German Research Foundation (DFG, project 504715884 and 235577387/GRK1974) is gratefully acknowledged. This research was also funded in part by the Austrian Science Fund (FWF) through grant 10.55776/FG6 TP5. Computational infrastructure and support were provided by the Centre for Information and Media Technology at Heinrich Heine University Düsseldorf. Fischer is also affiliated with DICE. Martin is also affiliated with DICE, CEPR, and CESifo. Schmidt-Dengler is also affiliated with WIFO, CEPR, and CESifo.

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Journal of the European Economic Association 2025 0(0):1–45

<https://doi.org/10.1093/jeaa/jvaf061>

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Teaching Slides

A set of Teaching Slides to accompany this article is available online as Supplementary Data.

1. Introduction

How changes in market structure affect competitive outcomes and consumer rents is a long-standing question in economics. The overall welfare effect of an increase in the number of firms, that is, entry, in oligopoly settings is ambiguous, as entry involves fixed costs but generally lowers prices and improves product quality (Mankiw and Whinston 1986). Therefore consumers, on average, benefit from entry.

However, when not all consumers are equally informed about prices in the market, their gains of entry are potentially heterogeneous. To illustrate this idea, suppose that there are two types of consumers in a homogeneous product market (Varian 1980). *Uninformed* consumers do not compare prices and simply purchase from a firm at random. For these consumers, the expected transaction price equals the average price in the market. *Informed* consumers, in contrast, observe all prices. The average transaction price of such a consumer corresponds to the expected *minimum* price in the market instead. In the presence of price dispersion, the effect of entry for these two groups depends on how entry affects the price distribution.¹

Price dispersion is ubiquitous in markets for physically homogeneous goods to an extent that cannot be explained by differences in location, cost, or services. Thus, it is important to understand how entry affects the distribution of prices and, subsequently, how consumers holding different information about prices are heterogeneously affected.²

In this paper, we give an empirical answer. We investigate how information about prices mediates the benefits of entry to consumers in the German retail gasoline market. Using real-time price data for the universe of German gasoline stations over five years and exploiting several hundred time- and geo-coded station entries, we identify the effect of entry on the entire price distribution.

The retail gasoline market provides an ideal setting to tackle the question at hand. First, despite an arguably fairly homogeneous product, gasoline markets exhibit substantial price dispersion. In our sample, the average market-level price range is 2.7 ct/l (Eurocents per liter), corresponding to almost 40% of an average station's margin. Given that the average German household spends more than 1,000 euros

1. In the model of sales introduced by Varian (1980), average prices even rise upon an exogenous increase in the number of firms from N to $N + 1$, so that uninformed consumers are hurt by entry and only informed consumers benefit. Lach and Moraga-González (2017) show that this result does not extend to richer heterogeneity in consumer information.

2. As consumer information and search behavior are strongly correlated with market-specific socio-economic variables such as income (Nishida and Remer 2018; Byrne and Martin 2021), heterogeneous effects along the information distribution imply different welfare effects across socio-economic backgrounds.

p.a. on fuel (Federal Ministry for Economic Affairs and Energy 2021), distributional implications across heterogeneously informed consumers matter. Second, stations primarily compete on the price dimension where menu costs are effectively zero. Third, the retail gasoline industry consists of numerous small markets, in which entry by one more station is likely to have non-negligible price effects. The spatial nature of competition in combination with a lack of nationwide pricing allows us to find suitable control markets. Last, the several hundreds of entries into narrow, isolated markets enable us to study heterogeneous effects of entry depending on station and market characteristics.

Exploiting the staggered nature of entry events in a difference-in-differences event study design, we first quantify the effect of entry on incumbent stations' average prices as a benchmark. Following Arcidiacono et al. (2020), the entry effect can be identified under the assumption that the precise timing of entry is exogenous from the incumbent's perspective. Leveraging the detailed within-market variation in competition, we estimate that average prices decrease by around 7% of a station's margin due to entry. This effect is persistent for at least three years post-entry.

Next, we explore the effect of entry on the price distribution at different quantiles by implementing the recentered influence function (RIF) regression approach of Firpo, Fortin, and Lemieux (2009). To infer the counterfactual price distribution, we employ the distribution regression approach by Chernozhukov, Fernandez-Val, and Melly (2013). Entry reduces incumbents' prices at all quantiles of the price distribution. The post-entry price distribution is first-order stochastically dominated by the counterfactual distribution of prices absent entry. Hence, all consumers benefit, in expectation, from an increase in competition. However, entry affects the price distribution asymmetrically: The price decrease from entry is most pronounced at the left tail of the distribution, where prices decrease by around 1 ct/l, or 13% of stations' gross margins. At the top of the distribution, price changes due to entry are not significantly different from zero. As entry might additionally facilitate price comparisons due to more stations in the consumer's direct vicinity, our estimates are a conservative estimate of the gains from entry.

When consumers are differentially informed (i.e., they observe a different number of price draws before making a purchase decision), their expected *transaction prices* differ, even when they face the same *posted prices*. To quantify this heterogeneous effect, we focus on the value of information (VOI): the difference between the average and the minimum posted prices in a market. This metric captures the gains of being fully informed about prices relative to observing only one price at random. We show that entry increases VOI by 29%. Hence, better-informed consumers benefit considerably more from entry.

The estimated effects are robust to a variety of sensitivity checks. We show that entry persistently increases the number of firms in the treated markets. Entry neither fosters nor anticipates exit; that is, it does not take place in response to exit. Using hourly traffic data from almost 2,000 traffic counters in Germany, we also show that entry does not take place in response to a demand shift (Dubois, De Mouzon, and Scott-Morton 2015), that is, an increase in traffic in the market. Our estimates further show no

effect of entry on incumbents' opening hours. This suggests that incumbents' primary response to entry is price adjustment. Beyond that, we show that entry also does not affect consumer information frictions in the market. Lastly, our findings are robust to various specifications, alternative region-time-specific controls, different estimation methods, and the time of day analyzed. Finally, we carry out the analogous exercise to examine the effect of exit. The results on exit mirror the effects found for entry: Exit increases prices, and this effect is more pronounced at the lower end of the distribution.

The contributions of this paper are multifold. First, by causally uncovering the heterogeneous consumer gains from entry and competition, our paper informs policymakers about a new dimension in the welfare effects of competition policy. For example, merger control and entry regulation (indirectly) affect the number of competitors in a market. Our paper sheds light on the fact that the average price effects of policy measures can strongly differ from price effects for specific consumer groups.

We further contribute to debates on whether and how policymakers should provide information to consumers. In many industries, policymakers have launched price comparison tools, for example, for financial products or energy contracts. Also in the gasoline markets of, for example, Germany, France, Austria, or Western Australia, price comparison tools for consumers were introduced to reduce search costs. While most academic research focuses on the question of whether transparency policies foster firm coordination more than consumers' ability to find the lowest price (Kühn and Vives 1995; Ivaldi et al. 2003; Rossi and Chintagunta 2016; Luco 2019), our paper unveils a positive effect of such transparency policies: Marginal consumers who become better informed due to such tools benefit more from an increase in competition, *ceteris paribus*. Hence, transparency policies have different effects on (un)informed consumers depending on the competition intensity in a market.

Second, we contribute to research studying the effect of firm entry on prices. The largest body of research analyzes retail grocery and the airline industry. Basker (2005) and Basker and Noel (2009) show that Walmart's entry reduces competitor prices as well as market-level average prices. Arcidiacono et al. (2020) find that Walmart's entry substantially reduces revenues for incumbent grocery stores, but they do not identify a significant price response. Bauner and Wang (2019) and Oschmann (2022) show that incumbents instead react to Walmart or Costco entry by reducing product variety and adapting their pricing strategies to more fluctuating prices. Atkin, Faber, and Gonzalez-Navarro (2018) and Busso and Galiani (2019) study retail markets in Mexico and find price effects of entry in the range of 2%–6%. Bernardo (2018), Cardoso et al. (2022), and Davis, McRae, and Seira (2022), investigating gasoline markets in Spain, Brazil, and Mexico, respectively, similarly find a negative average price effect of entry like us, but do not analyze distributional effects.

For the airline industry, Whinston and Collins (1992) show that incumbent airlines lower prices on the entered routes. Goolsbee and Syverson (2008) establish that incumbents lower prices already in response to the mere threat of entry by a discount carrier. Prince and Simon (2015) show that this response is accompanied by a reduction in service quality as measured by the on-time performance of incumbents. Reiffen and Ward (2005) employ the timing of FDA approval as exogenous in the

pharmaceutical industry to establish how generic entry competes away price-cost margins. We complement this literature by looking at a novel information channel that shapes entry effects on consumers. Additionally, we provide new evidence on the average price effects of entry based on hundreds of entry events in narrowly defined markets in which regional and national pricing are less concerning.

Third, we add to the empirical literature on the interaction of competition, informational frictions, and price dispersion, especially in gasoline markets, such as Barron, Taylor, and Umbeck (2004), Chandra and Tappata (2011), Lach and Moraga-González (2017), Lewis (2008), and Pennerstorfer et al. (2020). Rather than relying on cross-sectional variation in the number of competitors between markets, our event study design exploits within-firm and within-market price variation and isolates the causal effect of entry. Moreover, the setting enables us to explicitly study how consumers are differently affected by changes in competition, depending on how informed they are about prices. While Barron, Taylor, and Umbeck (2004) and Lewis (2008) find a negative relation between competition and price dispersion, we find causal evidence for a positive relation, which is in line with Pennerstorfer et al. (2020).

Finally, our paper complements the few papers that explicitly examine the effect of competition on the whole price distribution. Allen, Clark, and Houde (2014) study a bank merger and its heterogeneous effect on mortgage prices across local markets. In contrast to us, they exploit the ownership structure in the market, but not changes in the number of outlets. Moreover, the entry of different types of stations into submarkets allows us to study how entrant and incumbent characteristics affect the change of the price distribution. Lach and Moraga-González (2017) explore how the number of competitors in local gasoline markets affects the price competition in cross-sectional analysis between markets. Similar to us, they find that competition shifts prices in a first-order stochastic manner. Contrary to us, in their paper, prices in the right tail of the distribution decrease the most with competition.

The remainder of the paper proceeds as follows: In Section 2, we discuss the institutional setting of our research as well as the data we use. We then proceed with discussing the empirical strategy in Section 3. In Section 4, we present our results on the effect of entry on the price distribution, we discuss channels that give rise to these results, and we present several robustness checks. Finally, we conclude in Section 5.

2. Industry Background and Data

We study the effect of entries into the German gasoline market from 2015 to 2020 by combining data from various sources.³

We obtain the universe of prices and detailed information for all gasoline stations (brand affiliation, address, exact location) over the entire sample period from

3. Our sample ends in early 2020, the beginning of the COVID-19 pandemic. We drop the period due to substantial demand shocks as well as a temporary VAT reduction in 2020. See Montag et al. (2023a) and Fischer, Martin, and Schmidt-Dengler (2024) for studies on this VAT reduction.

Tankerking (2025).⁴ The market is dominated by five major brands: Aral (BP), Shell, Total, Esso, and JET, which together operate almost half of all stations in Germany (see Figure B.1 for a detailed overview). Around 25% of stations are not affiliated with any brand and are therefore considered independent. Since early 2017, algorithmic pricing software has been increasingly utilized (Assad et al. 2024).⁵

Gasoline stations on highways are arguably competing for a different set of consumers (Bundeskartellamt 2011; Montag et al. 2023a; Martin 2024). We follow the literature and omit these stations from our analysis entirely.⁶ Throughout, we focus on diesel prices at 5 pm, which is when most consumers fuel their cars (Deutscher Bundestag 2018). Later we also provide results for E5 gasoline and other times of the day (see Section 4.7).

Table 1 provides summary statistics on the stations in our sample, 15,437 in total, where approximately 40% are open 24/7. Across all stations, the average posted price over the sample period is 117 ct/l. Prices at major brands (Total, Shell, Esso, and Aral) tend to be higher, with an average brand premium of around 1 ct/l relative to unbranded stations, and even around 2 ct/l relative to budget stations such as JET stations.⁷

Gross margins account for around 6% of the retail price. Similar to Assad et al. (2024), we use daily, diesel wholesale price data for nine price regions in Germany as published in the Oil Market Report by Argus Media (2025) to calculate gross margins.⁸ We match the closest available regional wholesale price to each station on a daily basis. The average margin of slightly more than 7 ct/l fits industry survey evidence of approximately 10 ct/l (Scope Investor Services 2021).

Using stations' address information and location data, we compute the number of competitors within a given radius (Pennerstorfer et al. 2020). The average station has 0.9 (2.5) competitors within a 1 km (2 km) radius, respectively. The distribution of

4. The same data source is also used, for example, in Montag, Sagimuldina, and Winter (2023b), Montag et al. (2023a), and Martin (2024). This database uses real-time data provided by the German Market Transparency Unit (MTU, in German Markttransparenzstelle für Kraftstoffe MTS-K), which is a sub-unit of the German competition authority (Bundeskartellamt). By regulation, gasoline stations are required to submit all price changes to the MTU "instantaneously" since December 2013. The introduction of this regulation sufficiently predates our sample period and should therefore be inconsequential for our subsequent analysis.

5. Although pricing software might affect the industry in many ways, we show in Section 4.5 that the entry effects we are interested in in this paper seem orthogonal to the adoption of AI software.

6. We identify highway stations in a two-step procedure: First, we check stations on the websites <https://www.raststaetten.de/alle-standorte> and <https://serways.de/standorte>, which together list all highway stations let out to tenants by *Tank & Rast*, the company in charge of all highway roadhouses (Bundeskartellamt 2011), in a cross-section of February 2023. We get the exact coordinates of these roadhouses and their stations and calculate distances to stations in our dataset. Stations with a distance of below 400 m to a roadhouse are coded as highway stations. In the second step, we manually check our results for errors. We examine the address list of all German stations to identify stations IDs that were not yet coded as highway stations or were falsely coded as such. We classify almost 500 station IDs as highway gas stations and drop them.

7. See Online Appendix Figure S.1 for a more detailed distribution of brand premia.

8. The gross margin is given by: $\text{margin}_{it} = \text{price}_{it} / (1 + \text{VAT}) - \text{wholesaleprice}_{it}$, where the wholesale price already includes the diesel excise tax.

TABLE 1. Station-level information.

	Station Cross-Section			Station×Date Panel		
	<i>N</i>	Mean	S.D.	<i>N</i>	Mean	S.D.
Station choices						
Price (ct/l)	15,437	117.08	3.118	25,515,289	116.83	10.274
TOTAL	849	117.70	2.810	1,507,106	117.43	10.297
SHELL	1,714	117.64	2.447	3,070,060	117.57	10.228
ESSO	1,069	117.11	2.306	1,920,473	117.13	10.398
ARAL	2,255	118.19	2.307	4,042,296	118.20	10.539
JET	692	115.84	2.500	1,198,634	115.55	10.033
Others	8,858	116.72	3.448	13,776,720	116.27	10.148
Gross margin (ct/l)	15,437	7.449	1.639	25,515,289	7.401	3.085
1[Open 24/7]	14,835	0.394	0.483	5,996,285	0.382	0.486
Station characteristics						
# Competitors 1 km radius	15,437	0.869	1.079	25,515,289	0.885	1.093
# Competitors 2 km radius	15,437	2.500	2.599	25,515,289	2.574	2.640
1[Big four]	15,437	0.381	0.486	25,515,289	0.413	0.492
1[JET]	15,437	0.047	0.211	25,515,289	0.048	0.215
1[Entrant]	15,437	0.046	0.210	25,515,289	0.024	0.154
1[Incumbent - 1 km radius]	15,437	0.035	0.183	25,515,289	0.034	0.181
1[Incumbent - 2 km radius]	15,437	0.086	0.280	25,515,289	0.086	0.280

Notes: To obtain station-level, cross-sectional variables, we averaged over the respective observations at the station level. Using daily data from early 2015 to early 2020, we observe approximately 1,850 dates in our sample.

market sizes across all market-date observations is illustrated in Figure B.2. According to a market delineation based on a 1 km (2 km) radius, 47% (25%) of markets are local monopolists, and 30% (21%) are duopolies, respectively. Consequently, the majority of stations face local competition, even within fairly small radii.

In order to examine whether entry events are related to local changes in demand, we collect data proxying local fuel demand from two sources: First, we have traffic flow data from almost 2,000 traffic counters in Germany operated by Bundesanstalt für Straßen- und Verkehrswesen (2025). Second, we have access to very fine-grained annual population data at the 1x1 km-grid-level from the RWI-GEO-GRID dataset (Breidenbach and Eilers 2018; RWI 2023). Both data sources allow us to capture local trends within narrowly defined gasoline markets and also across municipality borders.

Finally, we complement our data with annual, county-level variables from the Statistical Offices of Germany and the Federal States. This data allows us to control for different local trends in GDP per capita, income, unemployment, population, vehicles, and commuting over time (Statistical Offices of Germany and the Federal States 2025a,b,c,d,e,f). We also obtained data on the number of vehicles and commuters at the county level.

2.1. Price Dispersion

An important characteristic of retail gasoline markets is cross-sectional price dispersion, which is substantial even though gasoline is a fairly homogeneous good.

TABLE 2. Descriptives on prices and price dispersion at market-date level.

1 km radius		p10	p25	Mean	p75	p90	S.D.
Mean Price	$\bar{p}_{it}$	103.9	109.4	116.6	124.2	129.2	10.22
Min Price	p_{it}^{min}	102.9	108.9	115.7	122.9	127.9	10.09
Max Price	p_{it}^{max}	104.9	109.9	117.6	125.4	130.9	10.46
S.D.	$\sqrt{\frac{1}{N-1} \sum_{i=1}^N (p_{it} - \bar{p}_{mt})^2}$	0.000	0.479	1.095	1.414	2.646	1.186
Range	$p_{mt}^{max} - p_{mt}^{min}$	0.000	1.000	1.856	3.000	4.000	2.031
VOI	$\bar{p}_{mt} - p_{mt}^{min}$	0.000	0.250	0.897	1.250	2.000	0.985
2 km radius		p10	p25	Mean	p75	p90	S.D.
Mean Price	$\bar{p}_{it}$	104.2	109.6	116.6	124.2	129.1	10.18
Min Price	p_{it}^{min}	102.9	108.9	115.4	122.9	127.9	10.01
Max Price	p_{it}^{max}	104.9	110.9	118.1	125.9	130.9	10.49
S.D.	$\sqrt{\frac{1}{N-1} \sum_{i=1}^N (p_{it} - \bar{p}_{mt})^2}$	0.000	0.577	1.251	1.704	2.748	1.118
Range	$p_{mt}^{max} - p_{mt}^{min}$	0.000	1.000	2.667	4.000	6.000	2.472
VOI	$\bar{p}_{mt} - p_{mt}^{min}$	0.000	0.500	1.239	1.667	2.750	1.167

Notes: Dispersion measures are only calculated for market-date combinations with at least two firms. Markets around entrants not included. All values in ct/l. p_{it} denotes a station's i price on date t . p_{mt}^{min} , p_{mt}^{max} , and $\bar{p}_{mt}$ denote the minimum, maximum and mean price in market m on date t , respectively.

We report several measures of price dispersion at the market level, namely the sample standard deviation of prices (*S.D.*), the range of prices (*Range*), and the value of information (*VOI*), that is, the difference between the *average price* and the *minimum price*, in Table 2.

We define markets as circles of 1 km and 2 km radius around each station.⁹ Within a 2 km radius, the average price range is around 2.7 ct/l; that is, 2% of the sample average price or almost 40% of stations' average gross margin (Scope Investor Services 2021).¹⁰ The average VOI is 1.2 ct/l. This implies that consumers could gain considerably, depending on how well they are informed about the price distribution. Moreover, this illustrates that we should not only investigate the effect of entry on average prices, but also on the price distribution, since consumers with different access to information will be heterogeneously affected by entry and profit from changes at different points of the price distribution.

Importantly, the price differences between stations cannot be fully explained by time-invariant differences in station characteristics such as service quality or other measures of vertical differentiation. We show this by documenting how the price ranking between stations changes over time. We implement the rank reversal test proposed by Chandra and Tappata (2011). For any pair of stations A and B , forming a couple c , the test calculates the likelihood that the station which is the cheaper on most

9. Similar approaches and distances are used in Chandra and Tappata (2011), Moraga-González et al. (2017), Cabral, Schober, and Woll (2019), and Pennerstorfer et al. (2020). In our analyses, we drop markets where entrants are the focal station, as we naturally do not observe these markets before entry events.

10. In some markets, dispersion is 0 ct/l (see values at first decile). This is due to the fact that most markets in our sample are duopolies or triopolies, where firms may also set the same price.

days sets the higher price. If station *A* usually is the cheaper station, then the following formula gives the rank reversal measure:

$$rr_c = \frac{1}{T_c} \sum_{t=1}^{T_c} 1[p_{A,t} > p_{B,t}], \quad (1)$$

where T_c is the number of days on which both stations operate simultaneously, and $p_{A,t}$ and $p_{B,t}$ are the respective prices on day t .

The average rank reversal of pairs with a maximum distance of 1 km (2 km) is 0.105 (0.117). Also, rank reversals increase with the distance between stations. Figure B.3 and Table C.1 show that reversals are lowest for station pairs up to 250 m distance between the stations; that is, the likelihood of rank reversal decreases by 5.7 percentage points relative to station pairs between 0.25 and 5 km distance. Hence, there is substantial temporal price dispersion at the station-pair level, which cannot be explained by time-invariant differences between stations. Information frictions can explain the observed changes in price rankings.

2.2. Market Entry

In Germany, gasoline stations are not subject to any geographic restrictions, and gasoline stations can, in principle, enter almost everywhere. The administrative process is fairly straightforward and common practice, especially for large chains, which are predominant in Germany. As for any business facility, setting up a gasoline station requires a building permit and a business registration in the respective municipality. On top of that, gasoline stations need to satisfy some other requirements regarding, for example, groundwater safety, fire safety, and emissions laws. As such, entry restrictions are viewed as a limited barrier to entry. However, setting up the infrastructure for a new gasoline station is costly and time-consuming, often taking several months.

We define entry as the first time a station posts a price, which should be very accurate given the mandatory price reporting regulation.¹¹ Based on these entry dates, we allocate treatment to all stations in a 1 km (2 km) radius at the weekly level.

We record over 700 unique entry events in our sample period. Overall, almost 1,300 stations (i.e., approximately 9% of all non-entrants in our sample) experienced at least one entry event within a 2 km radius over our sample period. More than 500 stations experienced entry even within a 1 km radius. In Figure 1, we illustrate the geographic and temporal distribution of entry events.¹² Entry occurred all over the country and is fairly evenly distributed over time. Slightly fewer entries in later years might reflect

11. Stations that changed their brand affiliation are also listed with a new identifier. We do not count these as “entry events”. For that purpose, we ensure that there is no station of (almost) identical address. Specifically, we check for each “potential entrant” with a formerly existing station within 400 m distance whether this station nearby might be the same station under a different affiliation. We check all these cases manually to arrive at a final list of entries. Despite all efforts, we might erroneously classify incumbent stations as entrants. In that case, we are likely to underestimate the price effect of entry in absolute terms.

12. Figure B.4 provides the distance distribution between entrants and incumbents.

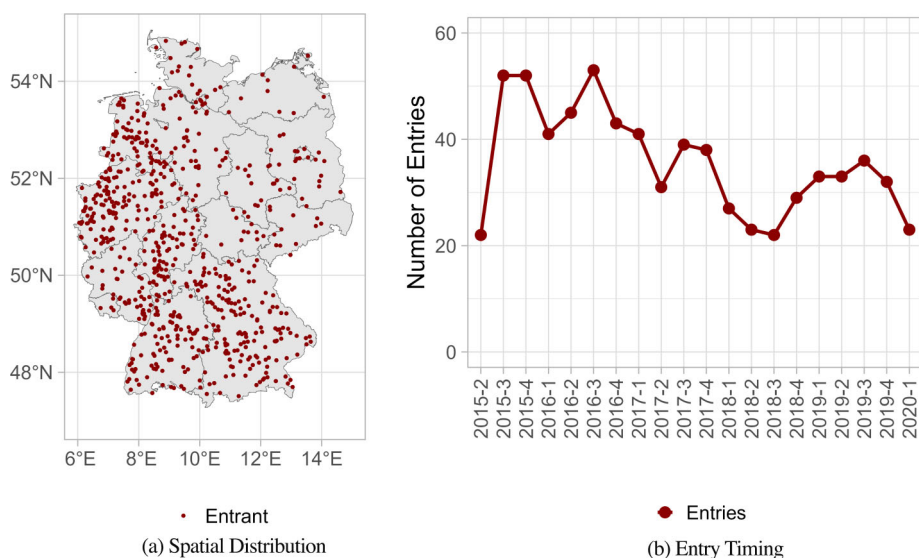


FIGURE 1. Geographical and temporal distribution of entry. The left figure documents the exact locations of all entry events in our sample. The borders represent the administrative federal state borders ($N = 16$ states). The shapefile is provided by ESRI Germany (2025). The right figure plots the distribution of entry events over time aggregated at the quarterly level.

the progressing consolidation of the industry as the number of stations in Germany has been falling over the last decades (Scope Investor Services 2021).

3. Empirical Strategy

We estimate the causal relationship between market entry and station- and market-level outcomes by exploiting the high number of market entry events documented above. In the following, we first explain the estimation setup. Second, we discuss how our identification strategy addresses standard endogeneity concerns when studying market entry. Lastly, we outline the estimation procedure, which allows us to study the heterogeneous effects of competition on consumer types, depending on their information about prices.

We study the effect of the staggered “rollout” of entries in a two-way fixed-effect difference-in-differences model, which compares prices of treated *incumbents* before and after entry with non-treated control stations. The baseline regression is as follows:

$$Y_{it} = \alpha_i + \lambda_{st} + \beta \times 1[\text{Post-Entry}]_{it} + \gamma' X_{ct} + \varepsilon_{it}. \quad (2)$$

Here, Y_{it} is station or market i 's outcome (e.g., station price or market mean price) on day t . Individual station (or market) and state-times-day fixed effects ($N = 16$ states) are denoted by α_i and λ_{st} , respectively. The vector X_{ct} , with c

denoting a county, consists of socio-economic variables at the county-year level (e.g., population, unemployment rate, and income), which we include as control variables. Most importantly, $1[Post-Entry]_{it}$ is the binary treatment indicator. It will equal one when an incumbent has experienced entry (within 1 km or 2 km radius, respectively) in period t or earlier. The error term is denoted by ε_{it} .

We only consider the first entry event for each incumbent and drop observations as soon as a second entry event occurs. By doing so, we avoid pooling the likely structurally different entry effects of a first and second entry. Further, we have a clear pre- and post-treatment period for each treated station then. This allows us to estimate the simple difference-in-differences coefficients β , which gives us the effect of entry on outcome Y_{it} , and enables us to implement the staggered difference-in-differences correction by Sun and Abraham (2021) for absorbing treatment on the same sample.¹³ We account for correlation in the residuals of the regression by clustering standard errors at the municipality level.

3.1. Identification

Analyzing firm entry is empirically challenging. Entry is part of firms' equilibrium behavior; that is, entry is endogenously determined by many market circumstances (see, e.g., Toivanen and Waterson 2005; Seim 2006; Berry and Reiss 2007; Schaumans and Verboven 2015; Bourreau, Sun, and Verboven 2021). The econometrician might not observe all market characteristics that determine entry incentives. Comparing *entrants with treated incumbents*, we show that entrants typically have significantly fewer competitors in a 1 km or 2 km radius even when controlling for narrow population data from the RWI-GEO-GRID data (see Table C.2). As profitability and prices are the main drivers of entry, a reverse causality bias is a concern. We address this problem by holding such unobserved characteristics of stations and markets fixed over time by including the respective fixed effects and by only exploiting the, from the perspective of the incumbent, exogenous timing of entry.

An additional challenge to identification may be that market-level profitability could change over time. This will be a problem if profitability does not change at the same time and in the same fashion in the treatment and control groups, that is, if treatment and control stations are not on the same trend. To alleviate this concern, we provide event studies for our difference-in-differences analyses estimated based on the following regression form:

$$Y_{it} = \alpha_i + \lambda_{st} + \sum_{\tau=-\bar{\tau}, \neq -1}^{\bar{\tau}} \beta_{\tau} 1[Entry]_{it,\tau} + \varepsilon_{it}, \quad (3)$$

13. Note that only 2.3% (1 km) and 4.6% (2 km) of all incumbents experience more than one entry event in our sample period. Only looking at the first four months after entry, no station in our sample experiences a second competitor entry.

where $\sum_{\tau=-\bar{\tau}, \neq -1}^{\bar{\tau}} 1[\text{Entry}]_{it, \tau}$ are leads and lags of the treatment. In our baseline regressions, we bin leads and lags to six-month bins with an effect window of four bins before and five bins after the treatment ($\underline{\tau} = \bar{\tau} = 4$).¹⁴ Endpoints are binned, that is, the most left and right bins include all observations lying outside the effect window.

If the pre-trends of the event studies are flat, this will be indicative of treated and control stations being on parallel trends. Simultaneously, it will alleviate concerns of a reverse causality bias. In the case of reverse causality, diverging pre-trends should be observable, too. We find that pre-trends are rather flat in all of our event studies later on. Comparing *treated incumbents with all other outside stations before treatment*, we even show that they are very similar in observable characteristics (see Table C.3). Incumbents do not differ in their price or margin level as well as county-level observables. Markets around incumbents only differ from non-incumbent markets in their higher local population level (3×3 km population data from the RWI-GEO-GRID data). This is suggestive of the estimated effects likely being close to representative of the overall population of stations within the same state-date cell in our sample, too.

Another concern is that we might pick up unrelated shocks, which occur at the same time and in the same markets as the entry events. Then, the estimated treatment effects would be unrelated to the entry events. However, the entry events are geographically dispersed and occur at different points in time. Hence, unrelated shocks would need to take place in a very similar nature in time and space as entry events to invalidate our identification strategy. This seems unlikely. To provide further support that no other demand-side factors change at the exact time of entry, we show that the traffic intensity does not change around the entry event. We use traffic flow data from almost 2,000 traffic counters all over Germany. The left panel of Figure B.5 shows that traffic at counters near entrants does not change around the time of entry.

Besides traffic, we study the development of local population numbers as a demand proxy. For this, we make use of annual 1×1 km grid population data from the RWI-GEO-GRID data. For each station, we determine in which grid cell it is located, and calculate the population in the nine grid cells around the station, that is, a grid of 3×3 km. This approach covers an area of 9 km^2 , which closely matches the area covered by our main estimation approach (3.14 km^2 and 12.56 km^2 for 1 km and 2 km markets, respectively). The right panel of Figure B.5 shows that the population does not change before or after entry in the area around entrant stations relative to areas around all other stations in Germany. This also holds for a wider 5×5 km grid to account for population changes in more distant areas.

Moreover, for our difference-in-differences regressions to uncover the causal relationship, it requires that the stable unit treatment variable assumption (SUTVA) is not violated. This implies that spillovers of the treatment of incumbents should not affect control stations. As identification stems from comparing prices within

14. We later on provide additional short-run analyses with weekly and monthly bins to study incumbents' immediate reactions to entry.

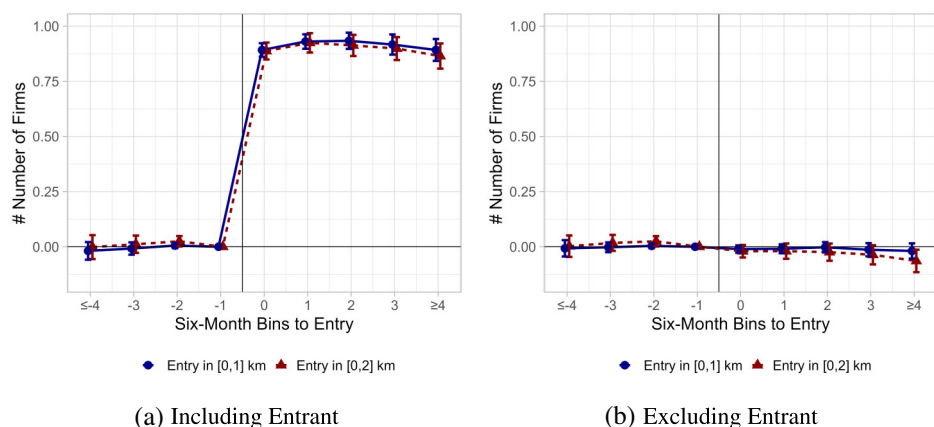


FIGURE 2. Market-level number of firms. This figure provides the leads and lags of the event study regression (3) with an effect window of four bins before and five bins after the entry event. The outcome variable is the number of firms in a market on a certain date. In Panel (a), entrants are included. In Panel (b), only incumbents are considered. Standard errors are clustered at the municipality level. Vertical bars indicate 95% confidence intervals. Endpoints are binned. County-level controls are included.

federal states and given the typically very narrow markets in the gasoline industry, it seems unlikely that control stations are strongly affected by entry. If at all, the strategic complementarity in incumbent and control station prices would result in underestimating the effect in absolute terms.

Furthermore, a natural concern is that the studied local shock on competition might not be persistent over time. For example, entry could occur in markets where exit has recently happened or is anticipated. Gasoline markets also have seen a decline in outlets across many countries throughout the last decades (Eckert and West 2005; Sen and Townley 2010).¹⁵ However, we provide evidence that the positive shock of entry on the number of firms is persistent. Figure 2 shows in its left panel that entry increases the number of stations in the short and long run for more than two years. That the number of firms does not jump up exactly by one is driven by the fact that some stations do not open every day. It is not driven by simultaneous exit; that is, we do not find that stations are triggered to exit the market in response to entry (right panel). Flat pre-trends also show that entry does not happen in markets that experienced a declining number of firms in advance.

Finally, the recent literature on two-way fixed effects difference-in-differences estimations with a staggered rollout of treatment shows that estimated effects might be biased (Callaway and Sant'Anna 2021; Sun and Abraham 2021; De Chaisemartin

15. The German gasoline market shrunk from 45,000 stations to less than 15,000 stations from 1970 to the early 2000s (Scope Investor Services 2021). However, since then the number of stations has remained relatively constant.

and d'Haultfoeuille 2020; Borusyak, Jaravel, and Spiess 2024). This will be the case if treatment effects are heterogeneous for incumbents who experience entry at different points in time. For example, due to changes in the search behavior of consumers over time, effect sizes might vary over time. Our results are unlikely to be qualitatively affected by this potential problem, as the vast majority of stations in our sample are never-treated control stations (Borusyak, Jaravel, and Spiess 2024). Nevertheless, we provide robustness checks with the proposed estimator of Sun and Abraham (2021) later on.

3.2. *Quantile Treatment Effects and Counterfactual Distribution*

We want to study how consumers, who hold different information about prices, are heterogeneously affected by station entry. Better-informed consumers purchase at different prices than less-informed consumers. We, therefore, examine how the price distribution changes over its entire range instead of just at the (conditional) mean. As we need to observe pre-entry prices to construct the counterfactual, this analysis focuses on the incumbent prices and does not include entrants' prices.¹⁶

First, we estimate the quantile treatment effects of entry. Formally, denoting the distribution of incumbents' observed post-entry prices by $F_{\text{entry}}(p_{it})$ and the counterfactual distribution of incumbents' prices absent entry by $F_{\text{no entry}}(p_{it})$, the quantile treatment effect at quantile q is defined by $F_{\text{entry}}^{-1}(q) - F_{\text{no entry}}^{-1}(q)$.

Note that, in our setting, we are especially interested in the *unconditional* quantile treatment effect instead of the *conditional* quantile treatment effect from standard quantile regressions. The latter gives the distributional effects conditional on covariates. For example, the conditional quantile treatment effect at the 90th percentile quantifies the effect on stations with relatively high prices (given their characteristics), but not on stations that are at the 90th percentile of the unconditional price distribution. We, however, care about the general effects along the price distribution—also across stations of different characteristics, as consumers likely compare such within a market—and hence are interested in the unconditional quantile treatment effects.

While the formal expression for the quantile treatment effect is rather simple, the econometrician's challenge is to obtain the unobserved counterfactual price distribution $F_{\text{no entry}}(p_{it})$. We implement two approaches that construct the counterfactual price distribution: First, we estimate quantile treatment effects by using the “RIF” by Firpo, Fortin, and Lemieux (2009). This method uses a local linear approximation of the counterfactual distribution based on observed prices to implicitly derive counterfactual prices. The approximation allows us to estimate quantile treatment effects with OLS regressions. While this provides valuable insights about the change at a chosen quantile of the distribution, it does not allow us to draw conclusions about the global shift of the price distribution. For example, testing

16. The latter would not contribute to the identification of the quantile treatment effects. However, they still matter for the consumer's choice and rents in the end, so that we will take them into account in our market-level analysis on market-level minimum, mean, and maximum prices later on.

whether the distribution of observed prices is first-order stochastically dominated by counterfactual (non-entry) prices is desirable, as this is a sufficient condition for all consumers to profit from entry in a homogeneous goods market. In a second step, we, therefore, elicit the *complete* counterfactual distribution of prices absent entry by adopting the “distribution regression” approach suggested by Chernozhukov, Fernandez-Val, and Melly (2013).¹⁷

Firpo, Fortin, and Lemieux (2009) relies on a local linear approximation of the counterfactual price distribution. The intuition behind the approach is as follows: Upon construction of a binary indicator of whether a price observation is above a price threshold p , a simple regression as in (2) yields the estimated vertical distance $F_{\text{entry}}(\hat{p}) - F_{\text{no entry}}(\hat{p})$ between the distribution of actual and counterfactual prices at a given price $\hat{p}$. The larger the empirical density at price $\hat{p}$, $f_{\text{entry}}(\hat{p})$, the smaller the change in prices (horizontal distance) needed to reach the found vertical distance. This horizontal price distance between the two distributions defines the quantile treatment effect. Formally, this is implemented by transforming the outcome variable to the “RIF”:

$$Y_{it}(\hat{q}) = p(\hat{q}) + \frac{\hat{q}}{f(p(\hat{q}))} - \frac{1[p_{it} < p(\hat{q})]}{f(p(\hat{q}))}. \quad (4)$$

Here, $Y_{it}(\hat{q})$ is the RIF and $p(\hat{q})$ the price at the $\hat{q}^{th}$ quantile. The function $f(p(\hat{q}))$ denotes the empirical density function of observed prices evaluated at $\hat{q}$. We estimate $f(p)$ using an Epanechnikov kernel with a bandwidth following Silverman’s rule of thumb as in, for example, Dube (2019). Given $\hat{q}$, $p(\hat{q}) + (\hat{q}/f(p(\hat{q})))$ is constant, so the main variation in the new outcome variable stems from whether the price is below or above $p(\hat{q})$. Estimating (3) with the RIF as the dependent variable allows us to obtain the quantile treatment effect at a certain quantile in only one regression. Applying this approach for different quantiles $\hat{q}$ allows us to estimate the shift at several discrete points of the entire price distribution.

Note that, as we insert (4) as the dependent variable in the regression equation (2), we identify changes in the empirical distribution within stations/markets over time.¹⁸

Chernozhukov, Fernandez-Val, and Melly (2013) propose to construct the counterfactual price distribution by estimating how entry affects the likelihood of prices being below a certain threshold. In our context, the respective regression equation is

$$1[p_{it} \leq \hat{p}] = \alpha_i + \lambda_{st} + \beta \times 1[\text{Post-Entry}]_{it} + \gamma' X_{ct} + \varepsilon_{it}. \quad (5)$$

17. The estimation of distributional effects of public policies is especially common in the field of public and labor economics. For example, see Havnes and Mogstad (2015) and Huebener, Kuger, and Marcus (2017) for applications of Firpo, Fortin, and Lemieux (2009). Dube (2019) and Hernæs (2020), among others, use the method of Chernozhukov, Fernandez-Val, and Melly (2013).

18. While many markets only have two or three stations, we can still obtain the quantile treatment effects at all quantiles, as identification stems from the price comparison of treated stations on any pre- to any post-entry date. State-date-specific demand or cost shocks are controlled for by the fixed effects.

The outcome variable is a dummy which will be one if $p_{it} \leq \hat{p}$. The coefficient β corresponds to the estimated vertical difference between the two distribution functions, $F_{\text{entry}}(\hat{p})$ and $F_{\text{no entry}}(\hat{p})$ at price $\hat{p}$. Estimating one regression for each unique price value in the dataset constructs the counterfactual empirical price distribution $\hat{F}_{\text{no entry}}(p_{it})$. Hence, this approach is computationally more intense but explicitly models the counterfactual distribution instead of a local linear approximation.

In our empirical analysis in Section 4, we implement both methods laid out here. We build on the approach by Firpo, Fortin, and Lemieux (2009) to estimate quantile treatment effects. Additionally, we estimate the price counterfactual distribution using Chernozhukov, Fernandez-Val, and Melly (2013) to then test for a global shift of the price distribution in a first-order stochastic manner, as elaborated in the following subsection. The outcome from Chernozhukov, Fernandez-Val, and Melly (2013) also serves as an input to cross-validate our results when using the method of Firpo, Fortin, and Lemieux (2009) instead, by comparing the empirical price distribution and the counterfactual price distribution.

3.3. First-Order Stochastic Dominance

The quantile treatment effects are, by design, informative about the effect of entry at certain points of the price distribution *in isolation*. However, we are also interested in how entry shifts the entire price distribution. For example, we can ascertain that consumers who are very well-informed about prices benefit from entry only if the price distribution is first-order stochastically dominated by the counterfactual (no entry) price distribution. This requires to *jointly* test for the difference between the distribution of observed and counterfactual prices at multiple prices.

We, therefore, implement three statistical tests to examine whether observed prices are jointly first-order stochastically dominated by counterfactual prices. The one-sided Kolmogorov–Smirnov (KS) test is based on the maximum vertical distance between two distributions. In contrast, the stochastic-dominance tests proposed by Barrett and Donald (2003) and Davidson and Duclos (2000) are based on the minimum vertical distance between two distributions and therefore more conservative. To compute those distances, the test by Davidson and Duclos (2000) takes a pre-specified set of points on the price grid as an input. Barrett and Donald (2003) integrate over the entire support of prices, which eliminates the necessity to fix a set of points upfront.

To examine the robustness of our findings, we present the results for all three tests. We compare observed post-entry prices with the counterfactual no-entry prices, elicited through the distribution regressions as in Chernozhukov, Fernandez-Val, and Melly (2013), as explained above.¹⁹ Note that the elicited counterfactual distribution,

19. An alternative would be to compare observed pre-entry and post-entry distributions. Since several additional determinants of prices, for example, input prices, might have changed meanwhile as well, this would be a potentially misleading comparison though.

given by the point estimates of the distribution regressions, may not be monotonically increasing at all prices. The distribution regressions at all prices do not require a positive density function across distribution regressions. However, this does not affect the test results, as the tests compare the observed price distribution and the counterfactual price distribution at certain prices only. As we take the point estimates of the estimated counterfactual distribution, we abstract from its confidence interval.

3.4. Effects on the VOI

The methods above allow us to estimate the changes along the price distribution. Importantly, this reflects only *posted prices*, which are different from the prices consumers actually pay (*expected transaction prices*). A useful concept to capture this disparity is the *VOI*, defined as the difference between the expected price and the expected *minimum* price in a market. VOI can then be interpreted as the extent to which fully informed consumers are better off relative to uninformed consumers purchasing from a station at random. An alternative interpretation is how much a yet-uninformed consumer would gain if she were to exercise effort to become fully informed (see Varian 1980; Armstrong, Vickers, and Zhou 2009; Lach and Moraga-González 2017; Honka, Hortaçsu, and Wildenbeest 2019, for a recent survey through the vast literature on consumer search).

To investigate this effect, we present event studies similar to our baseline specification above, using either VOI or the range of prices as the dependent variable.

4. Results

In this section, we first show that, indeed, prices *on average* decrease in response to entry. Building on this result, we then study the heterogeneous effects of entry along the entire price distribution and implications for consumers through the VOI.

4.1. Average Effect on Prices

In the first step, we estimate the dynamic effect of entry on posted prices at stations nearby as outlined in equation (3). We estimate the event studies for entry happening in a radius of 1 km or 2 km around a station. The left panel of Figure 3 shows that we find a negative effect of entry on prices. Flat pre-trends suggest that we do not violate the parallel trends assumption and, by that, are indicative of a causal relation between entry and prices. They are also suggestive of no anticipatory pricing in the months before entry because incumbents' prices right before entry do not significantly differ from prices more than two years before entry. The price effects are rather persistent over time for entry up to 2 km distance.

When entry occurs close by (i.e., up to 1 km distance between entrant and incumbent), the estimated average effect is a price reduction of around 0.5 ct/l. To

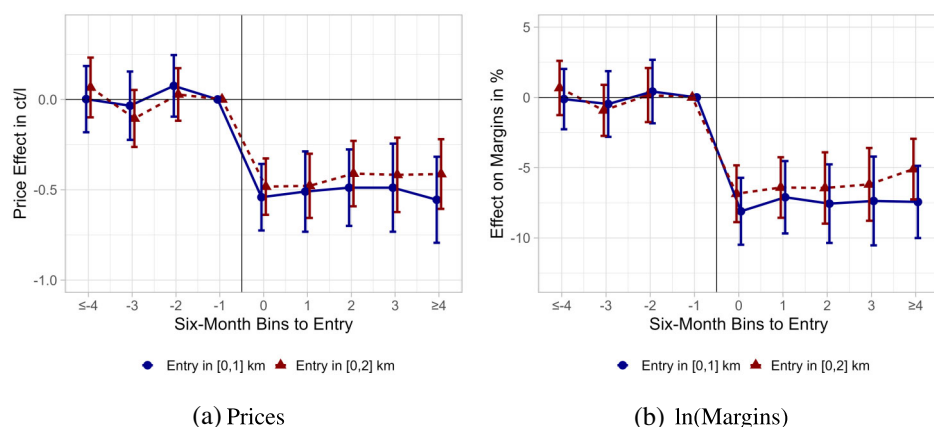


FIGURE 3. (Average) effect of entry on prices and $\ln(\text{margins})$. This figure presents the leads and lags of the event study regression (3) with an effect window of four bins before and five bins after the entry event. The outcome variables are prices in levels (Panel (a)) and logged margins (Panel (b)). Standard errors are clustered at the municipality level. Vertical bars indicate 95% confidence intervals. Endpoints are binned. County-level controls are included.

put these numbers into perspective, recall that the major brand premium is around 1 ct/l, so only around twice as big as the effect of a single entry. Moreover, a retail price reduction of 0.5 ct/l implies a reduction of around 7% of gasoline stations' gross margins. In the right panel of Figure 3, we provide event-study results on the effect on margins explicitly confirming this. In the cross-sectional study by Lach and Moraga-González (2017), a 10% increase in the number of stations is associated with a 0.06 ct/l price reduction in 1 km-radius markets. Since a single entry event increases the number of stations considerably in relative terms (e.g., by 50% and 20% for markets with two and five incumbent stations, respectively), our implied effects are much stronger than those in Lach and Moraga-González (2017).

As expected, the effect of entry is weaker for entry farther away from the incumbent. When pooling entry events in up to 2 km distance around incumbents, the price effect estimates are slightly smaller in absolute terms.

The respective pooled difference-in-differences estimates coinciding with the event studies can be found in Table C.3. In the Appendix, we also provide more detailed results on how the entry effect declines with increasing distance between entrant and incumbent (see Figure B.6).

4.2. Distributional Effects on Prices

We continue by analyzing the effect along the market-level price distribution. For this, we first show how market-level minimum, mean, and maximum prices are affected by

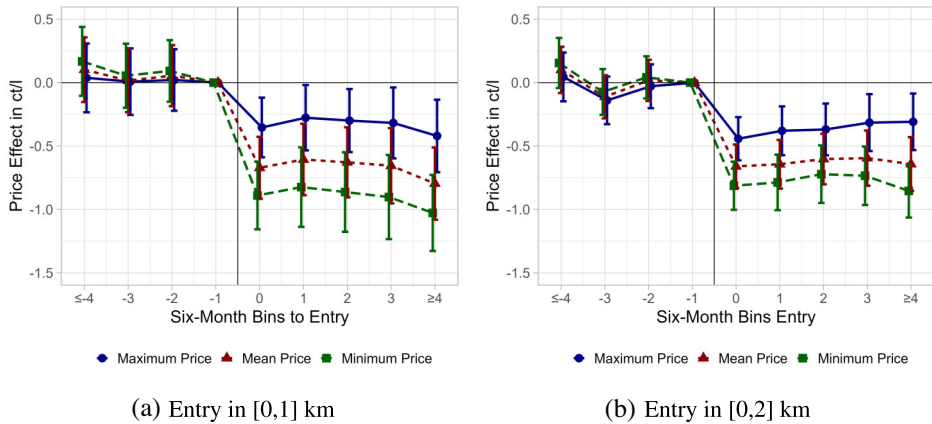


FIGURE 4. Effect of entry on the price distribution. This figure presents the leads and lags of the event study regression (3) with an effect window of four bins before and five bins after the entry event. Outcome variables are the market-level minimum, mean, and maximum price for each market-date observation. Panel (a) and (b) give results for different samples of relevant entry events. Only observations with at least two firms in the market included. Entrants' prices are included in the calculation. Standard errors are clustered at the municipality level. Vertical bars indicate 95% confidence intervals. Endpoints are binned. County-level controls are included.

the entry events. Subsequently, we investigate the effect of prices at all quantiles of the price distribution and estimate quantile treatment effects.

Figure 4 demonstrates that the minimum, mean, and maximum market prices are impacted differentially by entry.²⁰ For example, the maximum price decreases by less than 0.5 ct/l in response to entry up to a distance of 1 km. In contrast, the mean price drops by 0.7 ct/l, and the minimum price falls by 1.0 ct/l. The same qualitative pattern holds for markets of 2 km radius.

We draw the following conclusions from these findings: First, a decrease in the minimum price is indicative of fully informed consumers likely profiting more from entry. Similarly, a decreasing mean price suggests that uninformed consumers, who buy from a random station in the market, also benefit, albeit to a lesser extent than fully informed consumers. Second, the effect of entry on consumer prices is persistent over time.

To get an impression of how the price distribution changed at all prices instead of just the minimum, mean, or maximum price, we estimate quantile treatment effects as in Firpo, Fortin, and Lemieux (2009). We do this at every 5th percentile between the 10th and 90th percentiles. Figure 5 presents the estimation results. While the average price effect of entry in a 1 km radius around a station was approximately 0.5 ct/l, there is

20. Only market-date observations with at least two stations are included, since the minimum, mean, and maximum price are identical in markets with just one station.

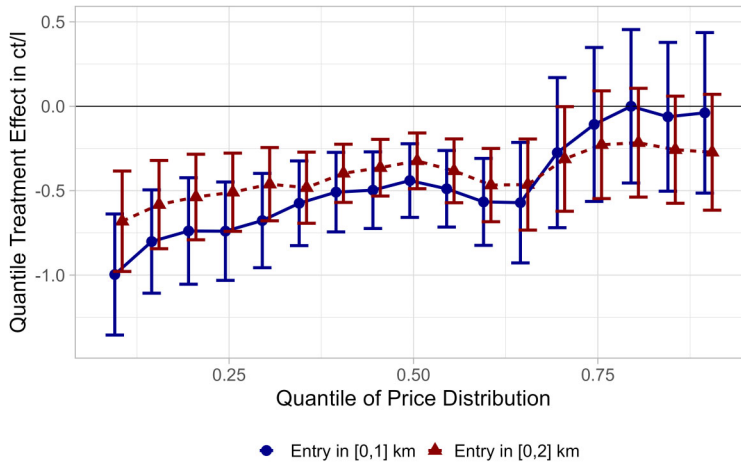


FIGURE 5. (Unconditional) quantile treatment effects of entry on prices. This figure plots quantile treatment effects of entry on prices for entry in 1 km radius (circle estimates) and 2 km radius (triangle estimates) estimated using the method of Firpo, Fortin, and Lemieux (2009). Estimates for every 5th percentile between the 10th and 90th percentile are provided. Standard errors are clustered at the municipality level. Vertical bars indicate 95% confidence intervals. County-level controls are included. Entrant observations are not included. Results obtained from regressions in which we also control for daily, region-specific wholesale price variations deliver almost identical estimates (see Figure B.7).

substantial variation in the price effect along the price distribution. Quantile treatment effects range from 1.0 ct/l at the left tail of the price distribution to 0.0 ct/l at the right tail of the distribution. For higher quantiles between the 65th and 85th percentiles, the effect is less pronounced. A similar pattern is observable for entry in a radius of 2 km, although the distributional heterogeneity is less pronounced.

Hence, prices at the left tail of the distribution respond more strongly to entry. This favors more informed consumers, as they are more likely to fuel at a low price. Nevertheless, uninformed consumers also profit from low prices, as the expected price decreases.

We cross-validate our results on heterogeneous price effects along the price distribution by eliciting the counterfactual price distribution as in Chernozhukov, Fernandez-Val, and Melly (2013). With the full counterfactual distribution at hand, we can inspect the difference between observed and counterfactual post-entry prices over the entire support. Figure 6 supports our findings that entry affects incumbents' prices, especially at the left tail of the distribution. Counterfactual prices (i.e., incumbents' prices after the timing of entry if entry had not happened) are significantly higher than observed prices at the left tail of the distribution.

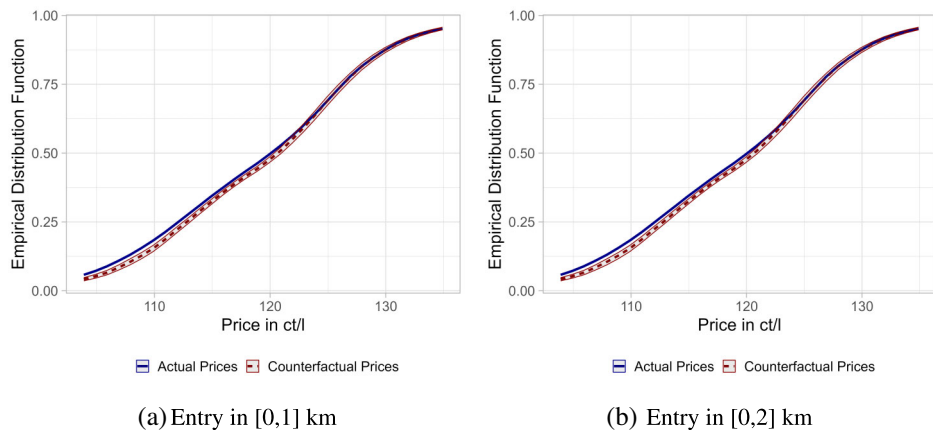


FIGURE 6. Observed prices and counterfactual distribution. The figures plot the actual price distribution of incumbents' prices, which experienced entry (solid line), and the estimated counterfactual price distribution (dashed line with confidence intervals) of a scenario without (or before) entry. Panel (a) and (b) give results for different samples of relevant entry events. The counterfactual distribution is obtained using distribution regressions as proposed by Chernozhukov, Fernandez-Val, and Melly (2013), that is, the regressions as in equation (2) for price thresholds at each integer ct/l. The estimated treatment effect per distribution regression is then added to the quantile of the empirical distribution function of actual prices at the respective threshold price. The figure is truncated at the 5th and 95th percentiles of the actual price distribution. Standard errors are clustered at the municipality level. Vertical bars indicate 95% confidence intervals. Endpoints are binned. County-level controls are included.

4.3. First-Order Stochastic Dominance

As Figure 6 shows, entry shifts the whole price distribution upward. If the price distribution is shifted upward at each percentile, this would be indicative of all consumers, independent of their information about prices, being positively affected by entry. We thus test whether observed post-entry prices are first-order stochastically dominated by counterfactual prices.

We provide results for three different tests: the one-sided KS test, as well as the two tests proposed by Davidson and Duclos (2000) and Barrett and Donald (2003). The one-sided KS test builds on the maximum, one-sided distance between the distribution of observed and counterfactual prices. The null hypothesis of equal distributions is tested against the alternative that the counterfactual distribution is smaller than the observed prices. However, for this, test only the maximum distance and not the distance between the distributions over the full support matters. The tests by Davidson and Duclos (2000) and Barrett and Donald (2003) examine whether the counterfactual distribution dominates the empirical distribution function of observed prices at multiple prices at the same time. In both tests, the null is first-order stochastic dominance. The alternative in Davidson and Duclos (2000) is no restriction, and in

TABLE 3. First-order stochastic dominance.

Entry in [0,1] km	H_0	Test-statistic	p -value (H_0 reject)
One-sided KS test	$F_{no\ entry} = F_{entry}$	0.0292	< 0.01
Davidson and Duclos (2000)	$F_{no\ entry} \leq F_{entry}$	$\in [-40.575, 0.915]$	–
Barrett and Donald (2003)	$F_{no\ entry} \leq F_{entry}$	0.195	0.93
Entry in [0,2] km	H_0	Test statistic	p -value (H_0 reject)
One-sided KS test	$F_{no\ entry} = F_{entry}$	0.0201	< 0.01
Davidson and Duclos (2000)	$F_{no\ entry} \leq F_{entry}$	$\in [-46.085, -8.633]$	–
Barrett and Donald (2003)	$F_{no\ entry} \leq F_{entry}$	–	1

Notes: This table reports the results of tests of first-order stochastic dominance. The tests differ in their design. The one-sided KS test tests the null hypothesis against the alternative that $F_{entry} > F_{no\ entry}$. It evaluates the maximum distance between the actual and counterfactual distribution. Davidson and Duclos (2000) extend the comparison of both distributions to more than the point of maximum distance. We implement the FOSD test by Davidson and Duclos (2000) as done in Asplund and Nocke (2006) and Gavazza (2011). Barrett and Donald (2003) smooth out the comparison by integrating over several points. Hence, the comparison can be interpreted as more continuous along the distributions. We evaluate the Davidson and Duclos (2000) approach at all prices in equal distance between 1.029 and 1.349 euro/l, which approximately represents the 5th and 95th percentiles of the distributions in steps of 2 ct/l (17 points). For Barrett and Donald (2003), we take the supremum of the distance between both ECDFs at every full ct/l between 1.029 and 1.349 euro/l. The critical value at a significance level of 0.01 for the Davidson and Duclos (2000) test comes from the studentized maximum modulus distribution (Stoline and Ury 1979). Given 17 equidistant points at which we calculate the test statistic, the critical value at the 1% significance value is 3.963. All test statistics have to be below this value, and at least one absolute value of a test statistic needs to exceed the critical value. No test statistic is reported for Barrett and Donald (2003), as the ECDF of counterfactual prices is smaller than the ECDF of observed prices at all examined prices, and thus the null hypothesis clearly cannot be rejected.

Barrett and Donald (2003), the observed prices first-order stochastically dominate the counterfactual prices.

Test results can be found in Table 3. The one-sided KS test rejects equal distributions for the one-sided alternative. Both other tests do not reject that the entry indeed shifts the price distribution in a first-order stochastic manner upward. The Davidson and Duclos (2000) test does not reject first-order stochastic dominance if all test statistics calculated at 17 equidistant prices between the 5th and 95th quantiles of the observed price distribution lie below the critical value of 3.963 (significance level 1%) and if at least one absolute value of a test statistic exceeds the critical value. Indeed, this is the case for our 1 km and 2 km markets. For the test of Barrett and Donald (2003), we report p -values of rejecting the null hypothesis based on comparisons of the two empirical CDFs (ECDFs) at every integer ct/l between the 5th and 95th quantiles of the observed price distribution. For both 1 km markets and 2 km markets, the null hypothesis of first-order stochastic dominance clearly cannot be rejected. For 1 km markets, the p -value is 0.93. For 2 km markets, the p -value is 1, since the counterfactual CDF is smaller than the observed CDF at literally all examined prices.

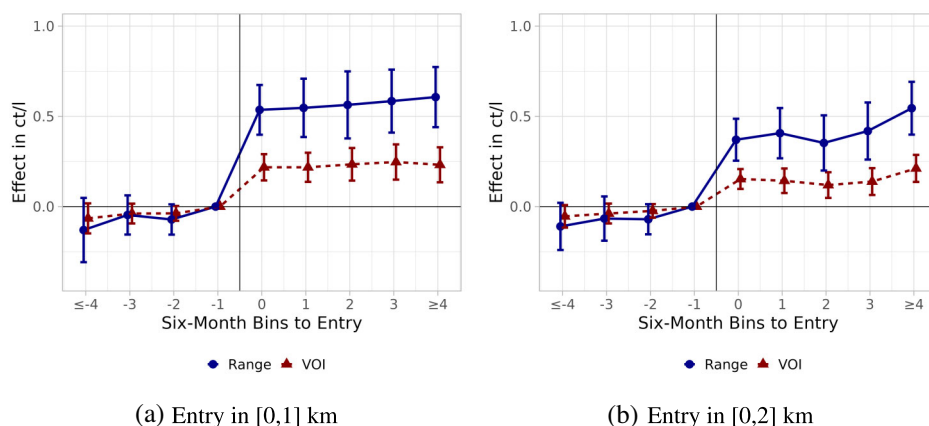


FIGURE 7. Market-level dispersion effects of entry. This figure provides the leads and lags of the event study regression (3) with an effect window of four bins before and five bins after the entry event. Panel (a) and (b) give results for different samples of relevant entry events. Only observations with at least two firms in the market included. Standard errors are clustered at the municipality level. Vertical bars indicate 95% confidence intervals. Endpoints are binned. County-level controls are included.

4.4. Effects on the VOI

In this subsection, we show how the VOI is affected by entry. This captures the saving potential for consumers through expected *transaction prices* instead of *posted prices*. Consumers with relatively easy access to information, or those who are very eager to engage in price comparison, will purchase at the expected *minimum*. The relative gain of doing so instead of purchasing at average prices is measured by VOI.

The effect of entry on the VOI is shown in Figure 7. The VOI increases by about 0.25 ct/l after entry in 1 km radius markets. Relative to the pre-entry average VOI, this corresponds to a 29% increase. The effect is persistent over time throughout our sample. The effect on the *range* of prices is even stronger, as it increases by 0.6–0.7 ct/l (i.e., by around 35%) after entry in 1 km radius markets. Consistently across measures and market delineations, we observe an increase in price dispersion. Post-entry, expected prices decrease, but expected *minimum* prices decrease even stronger, leading to an increase in VOI and the range.

Using quantile treatment effect methods as above, we further show that the increase in dispersion is evident along the entire dispersion distribution (see Figures B.8 and B.9). Hence, the VOI increases in all markets, although the effect is especially pronounced in formerly less dispersed markets. These findings are different from Moraga-González et al. (2017), who did not find increasing competition to be associated with higher dispersion. In contrast, Chandra and Tappata (2011) find more rival firms to increase market-level dispersion.

Importantly, our measure through VOI is likely to be a lower bound on the gains of entry for consumers. The reasons are the following. Throughout our main analysis, we

exclude the entrants' prices and consider incumbent prices only. However, consumers certainly do (weakly) gain additionally from the presence of an additional firm. Entry additionally effects consumers' propensity to search for lower prices. An additional supplier brings benefits for consumers, as it may provide easier access to more prices. This means that the effective number of prices observed by each consumer, for example during their daily commute, increases due to higher station density and reduced search costs (Dubois and Perrone 2017). The search incentives of consumers also change as an additional firm in the market affects the resulting price dispersion in the market. Given that we showed that price dispersion, and consequently, the incentive to search, increases with entry, both of these factors suggest that consumers are more inclined to compare prices. Absent a fully specified model about how the distribution of consumer information materializes, we cannot quantify this effect on transaction prices for different consumer types. We therefore take the rather conservative position that the number of price quotes observed by each consumer remains constant. In that sense, we are likely to underestimate the benefits of entry to consumers. Similarly, we abstract from aggregate demand effects due to lower prices. Although the demand elasticity in gasoline markets (Bento et al. 2009; Davis and Kilian 2011; Li et al. 2014; Coglianese et al. 2017; Kilian and Zhou 2024) or other real-time pricing markets (Fabra et al. 2021) is very low, this also contributes to us underestimating the overall welfare gains of consumers.

4.5. *Exit*

All our previous results so far examine the effects of station entry. It is instructive to see whether the effects are similar for exit. We therefore repeat the analysis for the mirroring scenario of station exit.

We find that an exit event increases incumbents' prices by 0.332 (0.336) ct/l in 1 km (2 km) radius markets (see [Online Appendix Figure S.2](#) in the Online Appendix for the dynamic effect). In line with our results on entry, we find that a change in competition affects prices the strongest in the left tail of the price distribution (see the right panel of [Online Appendix Figure S.2](#)). While the results for entry and exit are qualitatively identical, the effects of exit are slightly smaller in absolute terms. This stems from the fact that markets in which entry and exit take place are structurally different from each other. For example, there are on average 2.1 stations in a 2 km radius around entrant stations, but 3.8 stations around exiting stations. This suggests that additional effects, beyond the scope of this paper, arise under exit.

4.6. *Alternative Mechanisms and Heterogeneous Effects*

In this subsection, we first discuss and examine alternative channels via which stations could react to gasoline entry other than the pure price mechanism modeled above. We then examine whether the effect of entry is heterogeneous across different types of incumbents and entrants. The corresponding figures can be found in the Online Appendix.

Other Strategic Responses. We investigate whether stations react to entry by adapting non-price instruments such as opening hours in treated markets after entry. If opening hours, as a proxy of service quality and automation at the station, are not affected, this indicates that the firm's main responses to entry are price adjustments due to its low menu costs. [Online Appendix Figure S.3](#) shows how entry changes opening hours. To measure opening hours, we construct a daily station-level dummy that is one if a station opens 24/7. Incumbents' opening hours are not significantly different after entry in comparison to before entry. We take this as evidence that firms do not react by adjusting quality along this dimension (Armstrong and Chen 2009).

Finally, we examine another channel via which entry could have affected prices beyond the pure price pressure of consumers facing another alternative. In [Section A](#), we test how consumer information frictions change in response to entry by extending the test from Chandra and Tappata (2011) to a dynamic setting. Decreasing information frictions might also reduce incumbents' prices. We calculate quarterly station pair rank reversal measures for pairs of incumbent stations. In an event study setting similar to equation (3), we examine how rank reversals are affected when entry occurs near one of the stations of the couple. We do not find changes in information frictions (see [Appendix A](#)), which also suggests that the information distribution of consumer types does not change significantly with entry.

Price Cycles. Price cycles, reminiscent of the Edgeworth cycles (Maskin and Tirole 1988), are a well-documented feature in gasoline markets in many countries, such as the US (Lewis 2008), Canada (Noel 2007, 2009, 2019), Australia (Byrne and De Roos 2019), Norway (Foros and Steen 2013), and also Germany (Siekman 2017; Assad et al. 2024). Note that in our main analysis, we deliberately focus on daily prices at 5 pm to eliminate inter-temporal price effects generated through price cycles. To ensure that our results do not merely reflect an effect of entry on the price cycle, we first consider the frequency of price adjustment, measured by the absolute number of daily price changes per gasoline station, and the median price change, a measure of cycling asymmetry. The results are shown in [Online Appendix Figure S.4](#). In both of these measures, there is no entry effect, suggesting that entry did not alter the cycling behavior of stations in any significant way.²¹ Furthermore, in [Online Appendix Figure S.5](#), we demonstrate that the average effect of entry on prices and dispersion by hour does not vary significantly throughout the day. This evidence suggests that entry does not affect the nature of price cycles.

Algorithmic Pricing. According to Assad et al. (2024), many German gasoline stations have adopted algorithmic pricing software since early 2017, especially larger chains. They show that station prices are affected, especially when multiple stations in a market adopt the technology. To ensure that our results are not driven by the

21. Other studies like Noel (2007) and Siekman (2017) show that the existence of Edgeworth cycles or their characteristics depends on the local competitive environment. However, these studies compare cycles across markets or stations and do not exploit within-station changes in the competitive environment.

coincidence of entry and technology adoption, we separately analyze the period before early 2017, when almost no station had implemented pricing software yet. [Online Appendix Figure S.6](#) shows that both the average treatment effect and the distributional implications remain the same.

Heterogeneous Treatment Effects. We examine heterogeneity in the average and distributional effects of entry. We exploit the richness in the entry events across incumbent and entrant characteristics, for example, brand affiliation or exposure to local competition. [Online Appendix Figure S.7](#) shows heterogeneous effects on prices as estimated in equation (2). Note that, as before, we focus on incumbent prices throughout. We first consider heterogeneity regarding observable characteristics of the *entrant*. Prices decrease more strongly when low-price stations such as JET enter. We next investigate heterogeneity regarding observable characteristics of the *incumbent*. We find stronger price effects for brands that do not belong to the Big Four (Aral, Shell, Total, and Esso), and similarly, for formerly cheap stations. Both the entrant-specific and the incumbent-specific results are consistent with branded stations offering some additional amenity, such as faster service (Png and Reitman 1994), or are simply perceived as offering higher quality (Png and Reitman 1995; Martin 2024). These branded stations face less elastic demand and respond less aggressively to entry. Finally, there is a stronger price effect in markets in which the entrant brand did not operate a station before.²²

We further analyze whether the price effect of entry differs by the degree of automation of incumbent and entrant. Automation was linked to prices in gasoline markets outside of Germany (Kim 2018; Soeteven and Bružikas 2018). Note that our analysis on automation may be less relevant in Germany than in several other countries since Germany has a relatively low share of unmanned stations. As we do not observe the degree of automation of a certain station directly, we classify stations into automated and non-automated based on their brand affiliation and names. Some chains only operate automated, unmanned stations, and automated stations are named as such explicitly. We do not identify clear differences in the price reaction of automated and non-automated gasoline stations. Entry of automated and non-automated stations does not cause different price reactions either. Hence, no different implications for entry interacted with automation arise.

We show that the named dimensions of heterogeneity do not change the qualitative conclusions regarding the distributional effects, but the magnitude of these effects. Exemplarily, we look at the entry by the low-price brand JET.²³ We find that low-price entry substantially decreases prices in the left tail of the price distribution (see [Online Appendix Figure S.8](#)). At higher prices, quantile treatment effects are not statistically different from zero. This is also supported by the distributional analysis as

22. Entrants are of the same brand as incumbents in 2%–2.5% of all entry events (depending on whether markets are defined in 1 km or 2 km radii).

23. Prices by brand are reported in [Online Appendix Figure S.1](#).

in Chernozhukov, Fernandez-Val, and Melly (2013) (see [Online Appendix Figure S.9](#)). Correspondingly, the distributional implications of prices also hold for non-JET entries, but in a less pronounced way (see [Online Appendix Figures S.10](#) and [S.11](#)). We, further, show in [Online Appendix Figure S.12](#) that the market concentration determines how strong price reactions are at different quantiles. However, the pattern of stronger price effects at the left tail is robust across markets with a different number of non-entrant stations.

4.7. Robustness Checks

We continue in describing a set of sensitivity checks to prove the robustness of our estimation results.

Alternative Distributional Analysis. We study the heterogeneous effects of entry along the price distribution by applying an alternative approach to identify heterogeneous effects of a treatment along a distribution of an outcome variable—the method proposed in Cengiz et al. (2019). In separate regressions, we estimate how the treatment affects the likelihood of observing prices in certain price bins of the distribution. [Online Appendix Figure S.13](#) supports our results from above. Entry increases the probability of observing prices in the lowest price bins while reducing the probability of observing very high prices. Observing intermediate price bins is not significantly more likely than absent treatment.

Identification Cells and Inference. We test whether our results are robust to different identification cells, that is, identification within different layers of (non-)administrative regions. [Online Appendix Figure S.14](#) presents the baseline event studies for including Date, County-Date (401 counties), or ROR-Date (96 “Raumordnungsregionen”) fixed effects instead of State-Date (16 states) fixed effects.²⁴ A narrow identification grid reduces the average price effect of entry. Though the concern of spillovers within narrow geographical areas of counties might downward-bias the effect size (in absolute terms). The qualitative results hold for both market definitions. Also, the quantile treatment effects are similar for all fixed effects combinations, as evident in [Online Appendix Figure S.15](#).

We, moreover, check whether the interpretation of our results changes when standard errors account for correlation in residuals within stations, within counties, or within RORs instead of within municipalities. Our results are unaffected by this (see [Online Appendix Figures S.16](#) and [S.17](#)).

Binning. We show that the average estimated price effects are not sensitive to the choice of the number of leads and lags. [Online Appendix Figure S.18](#) shows that

24. We match stations to the different levels of regions using shapefiles by Bundesamt für Kartographie und Geodäsie (2025a,b).

the event studies look very alike for different approaches. We further report an event study (see [Online Appendix Figure S.19](#)), which zooms in on the weeks and months right around the entry events. This figure also illustrates that the main effect of entry materializes rather quickly, that is, within a two-week time window.

Market Size. To ensure that our effects are not just driven by very small, concentrated markets, in which entry might have the largest impact, we also estimate the distributional analysis for markets with at least four stations. In doing so, we focus on 2 km market only as the majority of 1 km markets have less than four active stations. Our results remain qualitatively the same (see [Online Appendix Figure S.20](#)).

Further, we restrict our baseline market-level analysis to markets with at least two stations. If entry into a monopoly market triggers exit, we would omit such markets as only one station remains. We, therefore, rerun our market-level analysis based on a sample where we include treated markets based on their pre-treatment concentration instead of by date-specific concentration. [Online Appendix Figure S.21](#) shows that our results remain unaffected if we only include treated markets with at least two stations in the market in the last pre-entry quarter.

Staggered Difference-in-Differences. Recent work by Borusyak, Jaravel, and Spiess (2024), Callaway and Sant’Anna (2021), De Chaisemartin and d’Haultfoeuille (2020), and Sun and Abraham (2021) has shown that treatment effect heterogeneity in combination with staggered treatment can bias the estimated treatment effect. In our setting, changes in search behavior over time might change the dynamic treatment effect across stations that are treated at different points in time. While the large number of never-treated units mitigates this concern (Borusyak, Jaravel, and Spiess 2024), we further address this concern by exemplarily also running our main regressions with the estimator proposed in Sun and Abraham (2021). The estimates support our qualitative findings. [Online Appendix Figure S.22](#) shows that entry (on average) decreases prices in a similar fashion as in Figure 3. If at all, the pattern becomes more extreme with no price effect at all for the top five deciles. Moreover, the right-hand side of the figure clearly shows that quantile treatment effects are stronger at the left tail of the price distribution. This also is in line with the differences between observed and counterfactual prices as reported in [Online Appendix Figure S.23](#).

Beyond that, treatment effects of entry may vary with the number of entries that take place near an incumbent. In our analyses, we decided to only focus on the first entry that an incumbent experiences (see Section 3). In [Online Appendix Figure S.24](#), we now further show that our results look identical when allowing for multiple treatments per incumbent so that the effects of consecutive entry events are treated as additive effects.

Alternative Daytimes and Types of Fuel. A concern in our main analysis might be that our results are sensitive to the time of the day or the type of fuel, for example, when the composition of consumers changes over the course of the day or demand

elasticity varies between fuel types (Montag, Sagimuldina, and Winter 2023b). Hence, we carry out the analysis also for 7 am and 12 am as well as for E5 gasoline prices instead of diesel. Online Appendix Figures S.25 and S.26 show, on the one hand, that results on diesel fuel are not strongly affected by the time of the day. On the other hand, there is no qualitative difference between our results for diesel and E5 gasoline.

5. Conclusion

In this paper, we study station entry in the German gasoline market and its impact on prices and the implications for different consumer types. Owing to our comprehensive data set with a very high number of unique entry events, we thus complement existing findings in the literature regarding the *average* effect of prices. The average effect of entry is a price reduction of around 0.5 ct/l. For the firms operating in that market, this reduction is certainly non-negligible, as it corresponds to a reduction in gross margins by 7%. Since entry also affects the gains of search and thus the distribution of consumer information types, this is likely to be a conservative estimate of the effect of entry.

Additionally, we find that entry decreases prices in a first-order stochastic dominance manner. However, price effects are stronger at the left tail of the distribution. These changes along the price distribution have heterogeneous effects on consumers, depending on how well informed they are about prices. We show that entry increases the VOI by 29%, implying that informed consumers benefit considerably more from entry. Uninformed consumers, who already pay higher prices to start with, do benefit from entry, but less so than other consumer types. Thus, our results have important implications for the distributional effects of competition on different types of consumers in markets with information frictions.

In this paper, we do not take a position regarding which segments of the population tend to fall into the informed or uninformed consumer category, respectively (Byrne and Martin 2021). Irrespective of whether high-income consumers have more or less elastic demand, they may be more inclined to search (Nishida and Remer 2018; Fischer, Martin, and Schmidt-Dengler 2024), rendering them the main beneficiaries of entry. In that case, policies regarding entry barriers and market structure also need to take into account distributional consequences. Through this channel, our findings inform policymakers to better understand the relevance of competition for different consumer groups.

Appendix A: Information Frictions

Price dispersion in homogeneous goods markets can arise from the fact that consumers hold heterogeneous information (Varian 1980). However, entry could improve consumer information about prices. More firms in a market imply a higher likelihood to observe prices (e.g., for commuters). Hence, entry may decrease information frictions.

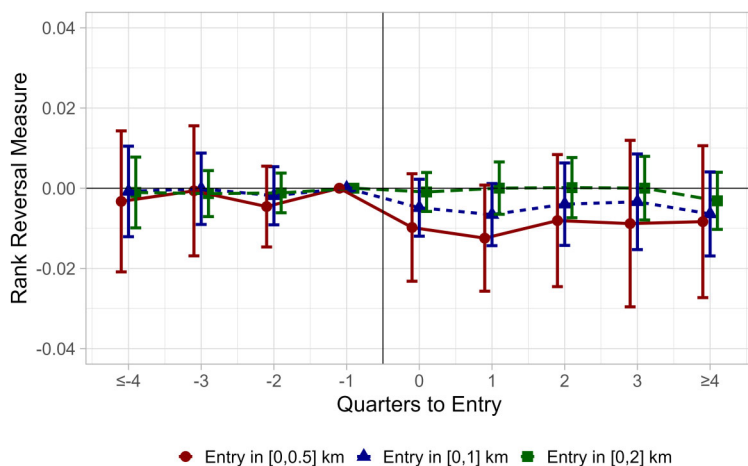


FIGURE A.1. Entry effect on rank reversal. This figure provides the leads and lags of the event study regression (A.2) with an effect window of four bins before and five bins after the entry event. The outcome variable is the rank reversal measure per couple and quarter constructed as in Chandra and Tappata (2011). The dataset includes all rank reversals of station pairs with a bilateral distance of 2 km maximum. This follows our market definitions from above. Standard errors are clustered at the municipality level. Vertical bars indicate 95% confidence intervals. Endpoints are binned. County-level controls are included.

Chandra and Tappata (2011) proposed a static test for information frictions based on how often the price ranking of a pair of neighboring stations alters. The intuition is that rank reversals in a homogeneous goods market can hardly be rationalized without information frictions. For a detailed explanation of the static test, see Section 2.

To examine how price rank reversals change over time and after entry, we extend their environment to a dynamic setting. Specifically, we calculate a rank reversal measure for each station pair and quarter. Hence, the outcome variable—the measure of rank reversal between station A and B (couple c) in quarter q where station A sets the lower price on at least half of all days in a quarter—is constructed in the following way:

$$rr_{c,q} = \frac{1}{T_{c,q}} \sum_{t=1}^{T_{c,q}} 1[p_{A,t} > p_{B,t}], \quad (\text{A.1})$$

where $T_{c,q}$ is the number of days in quarter q on which both stations report a price.

We then take $rr_{c,q}$ as the outcome in the following event-study regression:

$$rr_{c,q} = \alpha_c + \lambda_{sq} + \sum_{\tau=-\bar{\tau}, \neq -1}^{\bar{\tau}} \beta_{\tau} 1[\text{Entry}]_{c,q,\tau} + \varepsilon_{cq}, \quad (\text{A.2})$$

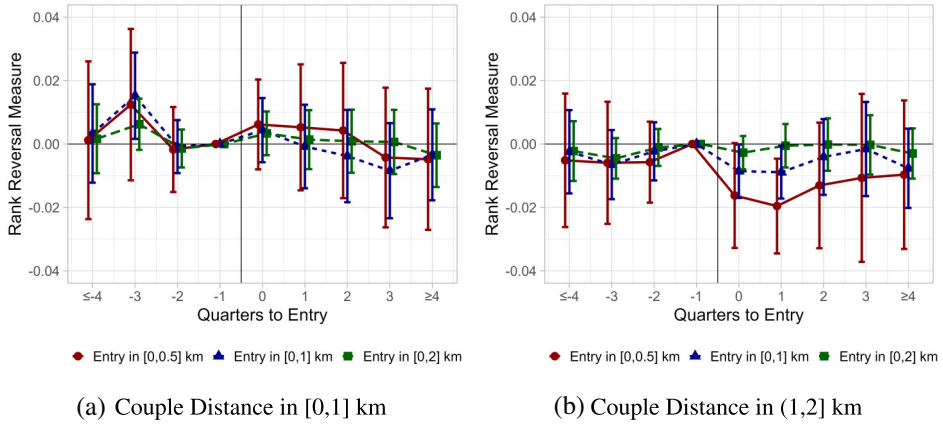


FIGURE A.2. Entry effect on rank reversal by couple distance. This figure provides the leads and lags of the event study regression (A.2) with an effect window of four bins before and five bins after the entry event. The outcome variable is the rank reversal measure per couple and quarter constructed as in Chandra and Tappata (2011). Panel (a) and (b) report the results of the entry effect for samples of different distances between stations. The dataset includes all rank reversals of station pairs with a bilateral distance of 2 km maximum. This follows our market definitions from above. The left plot looks at couples with a bilateral distance of 1 km maximum. The right plot looks at couples with a bilateral distance of between 1 km and 2 km. Standard errors are clustered at the municipality level. Vertical bars indicate 95% confidence intervals. Endpoints are binned. County-level controls are included.

where we include couple fixed effects (α_c) and state-quarter fixed effects (λ_{sq}).²⁵ $1[Entry]_{c,q,\tau}$ gives the leads and lags of entry events. A couple will be treated if at least one station experiences entry in a given radius. Similar to our approach in the main analysis, we only look at the first entry event per couple and drop observations after the date of the first entry.

Ex-ante the effect of entry on rank reversals is ambiguous. Entry might foster rank reversals because, for example, a new entrant very close to an incumbent might make the price ranking with other stations less important. At the same time, entry might improve consumers' knowledge about prices. Then, information frictions and rank reversals could decrease. Figure A.1 provides results of entry effects on rank

25. Note that for some couples, stations might not be located in the same federal state. Hence, we construct the point in the middle between both stations and match this point to federal states. A similar procedure allows us to match county-level covariates to the regression and allows for clustering at the municipality level.

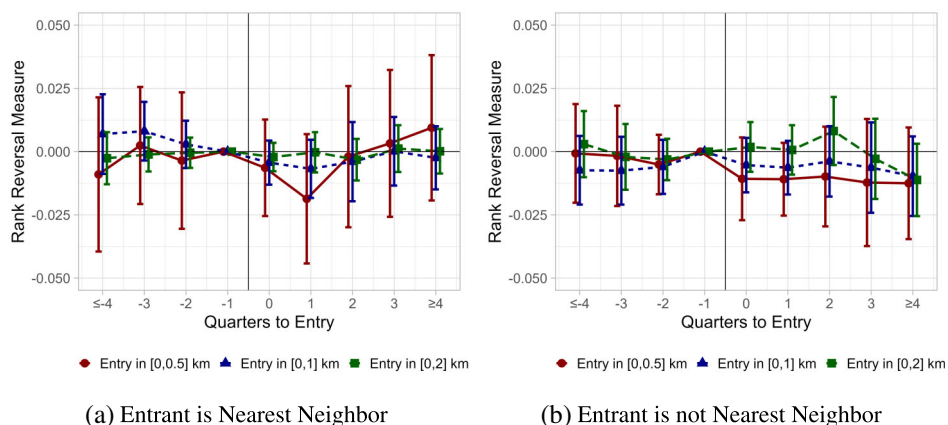


FIGURE A.3. Entry effect on rank reversal by entrant characteristic. This figure provides the leads and lags of the event study regression (A.2) with an effect window of four bins before and five bins after the entry event. The outcome variable is the rank reversal measure per couple and quarter constructed as in Chandra and Tappata (2011). Panel (a) and (b) report the results of the entry effect when the entrant is nearest (or not nearest) neighbor respectively. The dataset includes all rank reversals of station pairs with a bilateral distance of 2 km maximum. This follows our market definitions from above. The left plot looks at couples where the entrant is the nearest neighbor. The right plot looks at couples for which this is not the case. Standard errors are clustered at the municipality level. Vertical bars indicate 95% confidence intervals. Endpoints are binned. County-level controls are included.

reversals. We report the effect of entry in a 0.5 km, 1 km, and 2 km radius.²⁶ In all three cases, we cannot identify a change in the likelihood of rank reversal. Our results are also barely sensitive to whether looking at entry's effect on couples with a bilateral distance of up to or above 1 km (see Figure A.2) or only looking at entry events where the entrant is the nearest competitor of at least one of the stations (see Figure A.3). We take this as suggestive for entry not sufficiently affecting information frictions.

26. We add the more narrow entry radius of 0.5 km, as Chandra and Tappata (2011) suggest that rank reversal of very close stations is significantly lower than for other station pairs.

Appendix B: Additional Figures

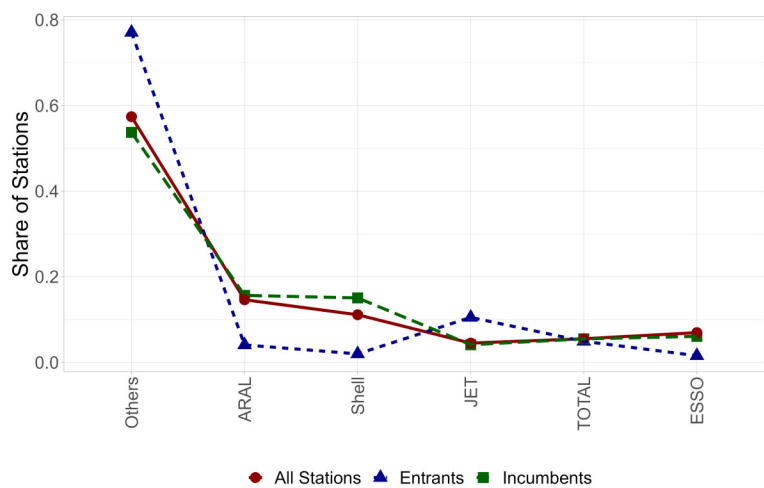


FIGURE B.1. Brand distribution. This figure shows the share of stations per brand among all German gasoline stations, for entrants only, and for incumbents only. Incumbents refer to stations in a 1 km radius around entrants. Other brands are pooled in “Others”.

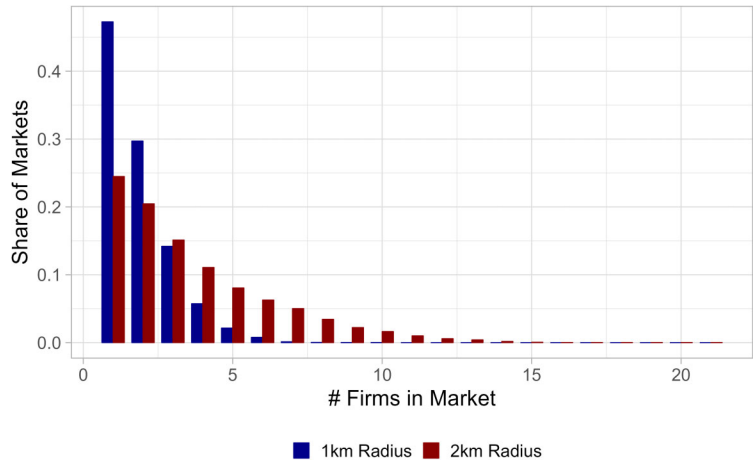


FIGURE B.2. Market size distribution—1 km and 2 km radius. The histogram plots the distribution of the number firms in a market using our baseline market definitions: Circles of 1 km or 2 km linear distance around each station. Markets around entrants are not included. Market size is defined at the market-date level.

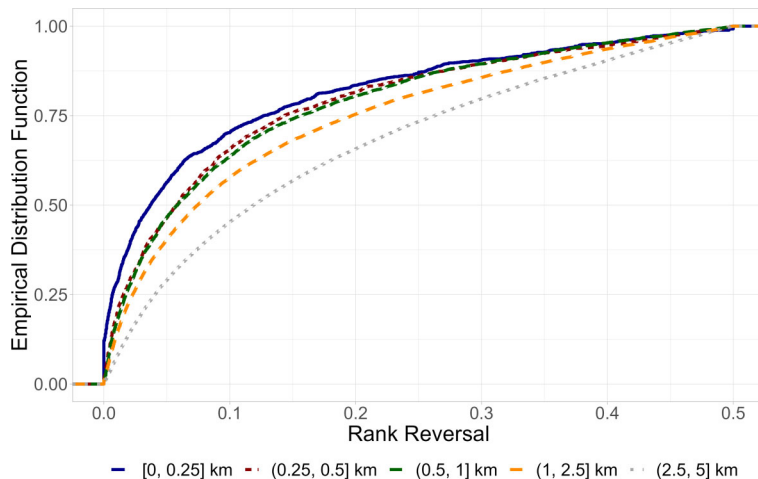


FIGURE B.3. Rank reversals—Chandra and Tappata (2011). Empirical distribution functions of station couples with heterogeneous distances between stations. The rank reversal measure follows the definition in Chandra and Tappata (2011): The share of observations, in which the mostly cheaper station, is the more expensive one.

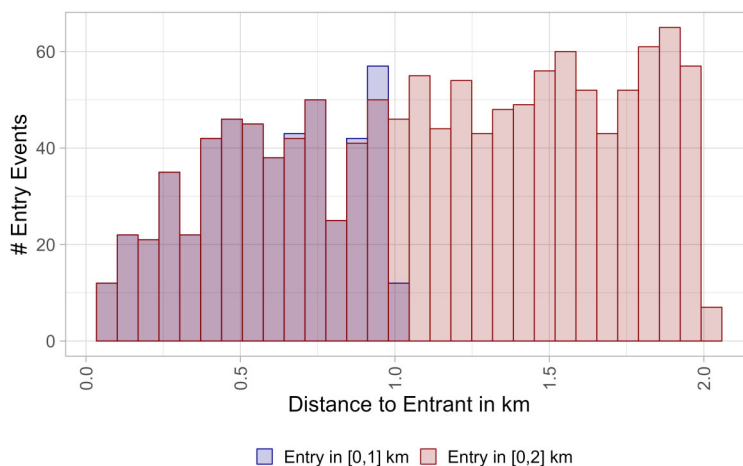


FIGURE B.4. Distance between incumbents and their entrants. Small differences between bins of entry in 1 km and 2 km radii for bins between 0 and 1 km distance stem from the fact that only the first entry is used as treatment in our baseline.

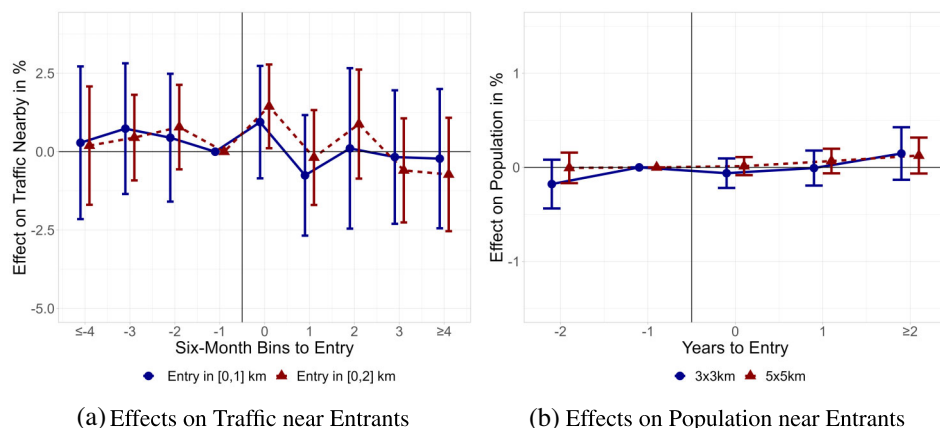


FIGURE B.5. Effect on demand proxies. The figures plot the effect of station entry on traffic flows between 4 pm and 6 pm at traffic counters in Germany (Panel (a)) as well as on annual population data in a 3×3 km grid around stations (Panel (b)). Traffic counters in less than 1 km or 2 km to entry are treated in this analysis. Standard errors are clustered at the municipality level. Vertical bars indicate 95% confidence intervals. Endpoints are binned. County-level controls are included. Observations after a second entry are not included in the traffic analysis. In the population analysis, entrants and the population in a 3×3 km grid around them are treated so that no multiple treatment occurs.

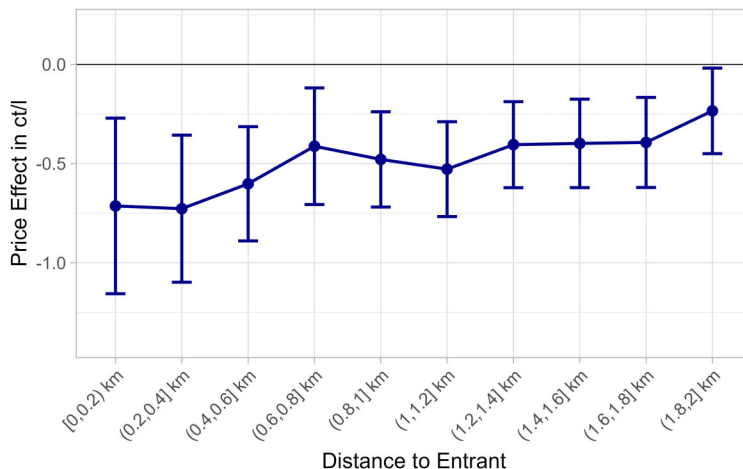


FIGURE B.6. Effect of entry on prices by distance to entry. This plot shows the effect of station entry on prices by distance between incumbent and entrant. Only the first entry event for incumbents within a 2 km radius is used. Standard errors are clustered at the municipality level. Vertical bars indicate 95% confidence intervals. Endpoints are binned. County-level controls are included. Observations after a second entry are not included.

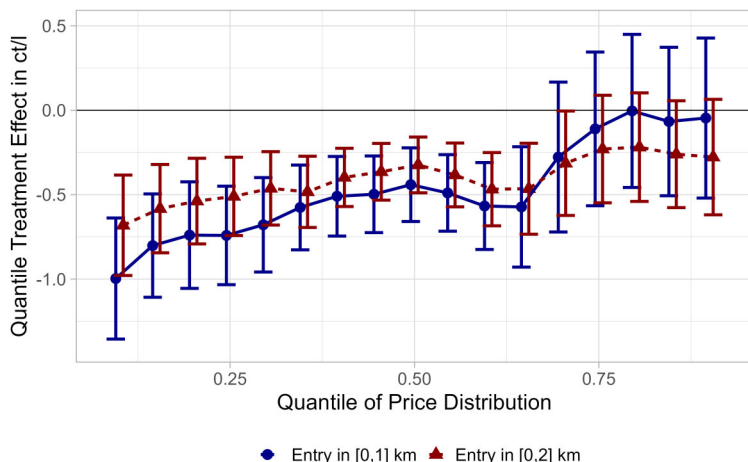


FIGURE B.7. QTEs of entry on prices—controlling for wholesale prices. This figure plots quantile treatment effects of entry on prices for entry in 1 km radius (solid line) and 2 km radius (dashed line with confidence intervals) estimated using the method of Firpo, Fortin, and Lemieux (2009). Estimates for every 5th percentile between the 10th and 90th percentiles are provided. Standard errors are clustered at the municipality level. Vertical bars indicate 95% confidence intervals. County-level controls are included. Results obtained from regressions in which we also control for daily, region-specific wholesale price variations.

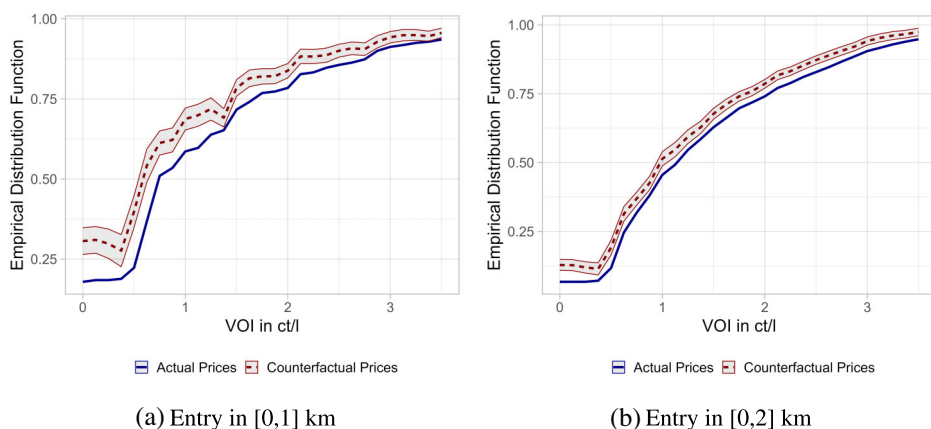


FIGURE B.8. Distributional effects of dispersion: VOI. The figures plot the actual price dispersion distribution of markets that experienced entry (solid line) and the estimated counterfactual price dispersion distribution (dashed line with confidence intervals) of a scenario without entry in the treated markets. Panel (a) and (b) present results for different samples of relevant entry events. The counterfactual distribution comes from distribution regressions as proposed by Chernozhukov, Fernandez-Val, and Melly (2013); that is, the regressions are in equation (2) for dispersion thresholds in 0.125 ct/l steps. The estimated treatment effect per distribution regression is then added to the quantile of the empirical distribution function of actual prices at the respective threshold price. The figure is truncated at the 5th and 95th percentiles of the actual price distribution. Standard errors are clustered at the municipality level. The shaded area indicates the 95% confidence interval.

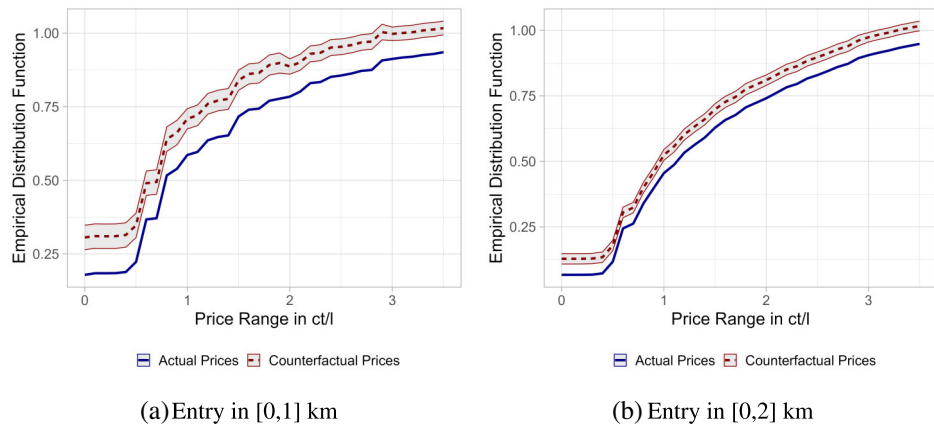


FIGURE B.9. Distributional effects of dispersion: Range. The figures plot the actual price dispersion distribution of markets, which experienced entry (solid line) and the estimated counterfactual price dispersion distribution (dashed line with confidence intervals) of a scenario without entry in the treated markets. Panel (a) and (b) present results for different samples of relevant entry events. The counterfactual distribution comes from distribution regressions as proposed by Chernozhukov, Fernandez-Val, and Melly (2013), that is, the regressions as in equation (2) for dispersion thresholds in 0.125 ct/l steps. The estimated treatment effect per distribution regression is then added to the quantile of the empirical distribution function of actual prices at the respective threshold price. The figure is truncated at the 5th and 95th percentiles of the actual price distribution. Standard errors are clustered at the municipality level. The shaded area indicates the 95% confidence interval.

Appendix C: Additional Tables

TABLE C.1. Rank reversals for different station pair distances.

	<i>rr_{AB}</i>	
	(1)	(2)
1[0 km ≤ distance ≤ 0.25 km]	−0.057*** (0.006)	
1[0.25 km < distance ≤ 0.5 km]		0.014** (0.006)
1[0.5 km < distance ≤ 1 km]		0.016*** (0.005)
1[1 km < distance ≤ 2.5 km]		0.035*** (0.005)
1[2.5 km < distance ≤ 5 km]		0.071*** (0.006)
Observations	81,415	81,415

Notes: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors are two-way clustered for the municipality of each station included in the couple. All station pairs of maximum distance 5 km are included.

TABLE C.2. Differences in levels between entrants and incumbents.

	1[Entrant] _{it}					
Market definition:	[0,1] km			[0,2] km		
Station-specific characteristics						
Price (ct/l)	−0.0184*** (0.0052)	−0.0180*** (0.0052)	−0.0083 (0.0055)	−0.0085* (0.0046)	−0.0080* (0.0046)	0.0033 (0.0043)
Gross margin (ct/l)	0.0028 (0.0060)	0.0029 (0.0060)	0.0100 (0.0065)	−0.0067 (0.0053)	−0.0063 (0.0052)	−0.0040 (0.0049)
# Competitors 1 km radius	−0.1700*** (0.0133)	−0.1607*** (0.0137)	−0.1485*** (0.0118)			
1[# Competitors 1 km > 0]×VOI		−0.0248*** (0.0054)	−0.0114*** (0.0058)			
# Competitors 2 km radius				−0.0420*** (0.0066)	−0.0391*** (0.0065)	−0.0384*** (0.0060)
1[# Competitors 2 km > 0]×VOI					−0.0208*** (0.0065)	−0.0143*** (0.0049)
log(Population - 3 × 3 km grid)	−0.0779*** (0.0127)	−0.0759*** (0.0126)	−0.0646*** (0.0125)	−0.1291*** (0.0131)	−0.1251*** (0.0130)	−0.0997*** (0.0122)
County-specific characteristics						
log(Unemployment rate)	0.0084 (0.0450)	0.0070 (0.0445)	0.0630 (0.0510)	−0.0225 (0.0378)	−0.0198 (0.0371)	0.0192 (0.0419)
log(Commuters)	0.0316 (0.0774)	0.0302 (0.0760)	0.0524 (0.0850)	−0.1088 (0.0673)	−0.1028 (0.0658)	−0.0522 (0.0677)
log(Population)	0.1787* (0.0955)	0.1788* (0.0942)	0.0705 (0.1065)	0.3414*** (0.0860)	0.3349*** (0.0851)	0.2635*** (0.0939)
log(Vehicles)	−0.2112* (0.1146)	−0.2096* (0.1136)	−0.1341 (0.1276)	−0.2153** (0.1024)	−0.2153** (0.1021)	−0.1910* (0.1120)
log(GDP p.c.)	0.0238 (0.0412)	0.0256 (0.0406)	0.0213 (0.0422)	0.0511 (0.0376)	0.0522 (0.0374)	0.0356 (0.0406)
log(Available income p.c.)	−0.0407 (0.1205)	−0.0302 (0.1184)	0.0532 (0.1350)	0.0229 (0.1174)	0.0325 (0.1164)	0.1017 (0.1198)
Brand FE	×	×	✓	×	×	✓
State-date FE	✓	✓	✓	✓	✓	✓
Observations	1,103,982			1,838,558		

Notes: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors are clustered at the municipality level. Linear probability model with dichotomous outcome. Only observations of entrants and incumbents after entry are included.

TABLE C.3. Differences in levels between incumbents and outsiders.

	1[Incumbent] _{it}					
Market definition:	[0,1] km			[0,2] km		
Station-specific characteristics						
Price (ct/l)	0.0008 (0.0006)	0.0008 (0.0006)	0.0006 (0.0006)	0.0005 (0.0013)	0.0005 (0.0014)	0.0002 (0.0013)
Gross margin (ct/l)	-0.0002 (0.0007)	-0.0002 (0.0007)	-0.0000 (0.0007)	0.0007 (0.0015)	0.0007 (0.0015)	0.0011 (0.0015)
# Competitors 1 km radius	0.0005 (0.0010)	0.0004 (0.0011)	0.0005 (0.0011)			
1[# Competitors 1 km > 0]×VOI		0.0001 (0.0008)	-0.0001 (0.0008)			
# Competitors 2 km radius				0.0025 (0.0017)	0.0024 (0.0018)	0.0023 (0.0018)
1[# Competitors 2 km > 0]×VOI					0.0010 (0.0014)	0.0005 (0.0013)
log(Population - 3 × 3 km grid)	0.0029*** (0.0010)	0.0029*** (0.0010)	0.0027*** (0.0011)	0.0097*** (0.0022)	0.0096*** (0.0022)	0.0092*** (0.0022)
County-specific characteristics						
log(Unemployment rate)	-0.0059 (0.0065)	-0.0059 (0.0066)	-0.0070 (0.0064)	-0.0130 (0.0146)	-0.0130 (0.0146)	-0.0156 (0.0145)
log(Commuters)	0.0038 (0.0095)	0.0038 (0.0095)	0.0030 (0.0096)	0.0075 (0.0247)	0.0076 (0.0246)	0.0066 (0.0244)
log(Population)	-0.0118 (0.0114)	-0.0118 (0.0114)	-0.0103 (0.0115)	-0.0316 (0.0286)	-0.0317 (0.0286)	-0.0283 (0.0284)
log(Vehicles)	0.0064 (0.0135)	0.0064 (0.0135)	0.0055 (0.0137)	0.0139 (0.0346)	0.0139 (0.0346)	0.0110 (0.0343)
log(GDP p.c.)	0.0008 (0.0052)	0.0008 (0.0052)	0.0007 (0.0052)	0.0004 (0.0128)	0.0004 (0.0128)	0.0007 (0.0128)
log(Available income p.c.)	-0.0290* (0.0153)	-0.0289* (0.0153)	-0.0286* (0.0149)	-0.0715* (0.0390)	-0.0715* (0.0390)	-0.0709* (0.0384)
Brand FE	×	×	✓	×	×	✓
State-date FE	✓	✓	✓	✓	✓	✓
Observations	24,411,307			23,676,731		

Notes: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors are clustered at the municipality level. Linear probability model with dichotomous outcome. Observations of entrants are dropped. Post-treatment observations of incumbents are dropped.

TABLE C.4. (Average) effect of entry on prices.

	$Price_{it}$	
	Entry in [0,1] km	Entry in [0,2] km
1[Post-Entry] _{it}	-0.536*** (0.084)	-0.450*** (0.063)
Station FE	✓	✓
State-date FE	✓	✓
Observations	24,887,154	24,860,948

Notes: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors are clustered at the municipality level. County-level controls are included. Observation numbers differ across treatment allocation, as observations of stations, which experience entry more than once, are dropped from the sample after the date of the second entry.

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Supplementary Data

Supplementary data are available at *JEEA* online.